Introduction to Software Engineering

Software Requirements Specification

ShiftManager – Team 6 (404NotFound) – PA2



Software Engineering Department
Faculty of Information and Technology
University of Science

Table of Contents

Objectives 1

1 Member Contribution Assessment 2

2 Introduction 3

2.1 Purpose and Scope 3

Purpose 3

Product Scope 3

2.2 Glossary 4

3 Problem Statement 5

Business Context and Problem Overview 5

Operating Environment 5

Design and Implementation Constraints 5

4 Stakeholders 7

5 Requirements 8

5.1 Functional Requirements 8

5.2 Non-Functional Requirements 11

5.3 Domain Requirements and Business Rules 12

5.4 Acceptance Criteria 15

6 Use Case Model 27

6.1 Use Case Diagram 27

6.2 Actors 27

6.3 Use Case List 27

7 Use Case Specifications 29

7.1 Use Case UC-01 29

7.2 Use Case UC-02 31

7.3 Use Case UC-03 33

7.4 Use Case UC-04 36
7.5 Use Case UC-05 38
7.6 Use Case UC-06 41
7.7 Use Case UC-07 44
7.8 Use Case UC-08 46
7.9 Use Case UC-09 48
8 Prototype / Mockups 52
9 Requirements Traceability Matrix 58
10 Requirements Review 60
11 AI Usage Log 64

Software Requirements Specification

Objectives

This document covers the ShiftManager software requirements and their verification.

- ✓ Completing the Software Requirements Specification (SRS) document with the following sections:
  - Problem Statement
  - Stakeholders
  - Functional, non-functional and domain requirements
  - Use case model and use case specifications
  - Prototype / mockups
  - Traceability and requirements review
- ✓ Including an AI usage log for this phase
- ✓ Understanding the Software Requirements Specification (SRS) document.

1 Member Contribution Assessment

ID	Name	Role(s)	Contribution (%)	Evidence
24127378	Nguyễn Huy Hoàng	Team Leader / Requirements Reviewer. UC-05, UC-09; Section 5.4 Acceptance Criteria; Section 9 Traceability Matrix; Section 10 Requirements Review; Section 11 AI Usage Log; Section 1 contribution summary; final PA2 consistency review and integration.	34%	https://github.com/gloryh241106/ShiftManager/issues/6
24127085	Võ Quang Minh	UI/UX & NFR Analyst. Section 4 Stakeholders; Section 5.2 Non-Functional Requirements (reassigned); Section 8 Mockups; UC-01, UC-02 (reassigned), UC-06, UC-07.	33%	PR #4 — interactive prototype
24127499	Lê Hoàng Phúc	Business Analyst. Sections 2–3; Section 5.1 Functional Requirements; Section 5.3 Domain Requirements and Business Rules (reassigned); Section 6 Use Case Model; UC-03, UC-04 (reassigned), UC-08.	33%	https://github.com/gloryh241106/ShiftManager/issues/7
24127568	Nguyễn Cao Trí	Team member. Original PA2 tasks were reassigned to Hoàng, Minh and Phúc.	0%	No completed PA2 deliverable recorded.

2 Introduction

2.1 Purpose and Scope

Purpose

This Software Requirements Specification (SRS) document defines the complete functional, non-functional, domain requirements, and use case specifications for the ShiftManager web application. It serves as the authoritative single source of truth for architectural design (PA3), test planning (PA4), software implementation (PA5), and final academic evaluation.

Product Scope

In-Scope

The application focuses strictly on automated shift scheduling and absence coverage coordination for a single factory production team. The MVP baseline consists of four core Must-have features (F01–F04):

1. F01 – Authentication, Role-Based Access, and Employee Account Management: The system authenticates users through Supabase Auth and enforces permissions for Employee, Manager, and Boss roles. Authorized Managers can create and manage Employee accounts within their assigned production teams. Initial privileged accounts are provisioned separately, and public self-registration is excluded from the MVP.
2. F02 - Constraint-Based Automatic Shift Scheduling: Supports voluntary overtime sign-up and Manager approval. Generates draft weekly rosters based on employee qualifications, availability, non-overlapping shift constraints, rest-time rules, and a 5-week workload fairness balancing window, allowing Manager review and publication.
3. F03 - Centralized Schedule & Change Acknowledgement: Presents published rosters, highlights assignment updates (+NEW, CHANGED, CANCELLED), and records explicit employee schedule view acknowledgements.
4. F04 - Absence Reporting & Shift Coverage: Enables employees to submit absence reports, allows Managers to record unannounced no-shows, identifies eligible replacement candidates, and completes shift transfers safely via database transactions.

Out-of-Scope

The following capabilities are explicitly excluded from the current project scope:

- Time Tracking & Attendance: Clock-in/clock-out functionality, timesheets, and tardiness logging.

- Payroll & Compensation: Wage calculation, overtime pay processing, or financial reporting.
- Full Leave Management: Vacation request workflows, paid leave tracking, or complex recurring availability forms.
- Peer-to-Peer Shift Swaps. Shift transfers in this MVP are limited to Manager-coordinated absence and no-show coverage after candidate acceptance and Manager confirmation.
- Multi-Department Operations: Managing multiple production lines, factories, or cross-department workforce transfers.
- Push Notifications & External HR Integrations: Mobile push notifications and integrations with third-party HR or accounting systems.

Intended Readers: The 404NotFound development team members (Nguyễn Huy Hoàng, Võ Quang Minh, Lê Hoàng Phúc, Nguyễn Cao Trí), Course Lecturers, and Teaching Assistants (TAs).

2.2 Glossary

Term	Definition
ShiftManager	A lightweight web application designed to automate shift scheduling and absence coverage for factory production teams.
Shift	A fixed working block within a day
Draft Schedule	An unpublished weekly roster generated automatically or adjusted manually by the Manager.
Official Schedule	A finalized weekly roster approved and published by the Manager, forming the binding schedule version.
Hard Constraint	A mandatory rule that must not be violated
Soft Constraint	An optimization target
Absence Report	A formal notification submitted by an Employee indicating inability to attend a future assigned shift.
Replacement Offer	An offer issued by a Manager to an eligible candidate to cover an open staffing shortage.
Acknowledgement	An explicit confirmation submitted by an Employee indicating that they have reviewed their latest published schedule updates.
Boss	A factory supervisor/owner role possessing read-only privileges to inspect team-wide staffing coverage and workload distribution.

3 Problem Statement

Business Context and Problem Overview

ShiftManager targets the shift scheduling and operational coordination challenges of a single packaging and production team within an industrial park facility. Production managers currently handle shift rosters using Microsoft Excel spreadsheets and communicate schedule adjustments through Zalo group chats. As worker headcount increases, this manual workflow creates four critical operational friction points:

1. Scattered Schedule Information: Workers must manually browse chat histories to confirm shift assignments, leading to confusion when spreadsheet versions clash with chat announcements.
2. Uncommunicated Schedule Changes: Last-minute schedule revisions fail to reach all affected employees, leaving managers unable to verify who has received updated rosters.
3. Unexpected Absences and Staffing Shortages: Workers report absences on short notice or fail to show up (no-show), forcing managers to manually search for available substitutes under severe time constraints.
4. Manual Scheduling & Perceived Unfairness: Hand-crafting rosters via spreadsheets introduces human error such as shift overlaps, rest-time violations, and uneven distribution of night shifts or total working hours.

ShiftManager provides a centralized web application that automatically generates draft rosters under explicit constraint logic, centralizes official schedule publication, tracks employee schedule acknowledgements, and streamlines the absence replacement workflow.

Operating Environment

- Client-Side: Responsive Web Application accessible across modern desktop and mobile web browsers supporting HTML5, CSS3, and JavaScript. Built using React.
- Server-Side: Node.js with Express framework providing a RESTful API layer with Role-Based Access Control (RBAC).
- Database & Auth: Supabase (PostgreSQL) for relational data persistence and Supabase Auth for identity management.

Design and Implementation Constraints

1. Role-Based Access and Account Provisioning: The system supports Employee, Manager, and Boss roles. Initial Manager and Boss demonstration accounts are securely provisioned before system use. Authorized Managers may subsequently create and manage Employee

accounts within their assigned production teams through UC-02. Public self-registration and unauthorized role elevation are not supported.

2. Scheduling & Coverage Logic:
    - Shift assignments must strictly comply with employee qualification and availability constraints.
    - Assignments must not overlap and must satisfy mandatory rest periods between consecutive shifts.
    - Draft rosters generated by automated solver logic require explicit Manager review and publication before becoming official.
    - Replacement assignments require explicit substitute acceptance followed by final Manager confirmation.
3. Data Consistency: Replacement confirmation and roster updates must execute within an atomic database transaction.
4. Process Constraints: Development strictly follows course waterfall phase gates, Conventional Commits formatting (feat:, fix:, docs:, test:), and protected main branch rules requiring peer code review.

4 Stakeholders

#	Stakeholder	Description	Interests / concerns
1	Factory owner / Production supervisor (Boss)	Oversees staffing and workload for the assigned production team through a read-only view. Does not edit schedules.	Visibility of staffing gaps; fair distribution of hours and night shifts.
2	Shift manager (Manager)	Manages one production team: creates and maintains Employee accounts, configures staffing requirements, generates and publishes weekly schedules, monitors acknowledgements, and coordinates absence coverage.	Less manual scheduling work; no conflicts or overlaps; enough staff on every shift; knowing who has seen an update.
3	Production worker (Employee)	Works the assigned shifts: views the schedule, acknowledges updates, reports absences, and responds to replacement offers.	A clear, up-to-date schedule; early notice of changes; fair shift assignments; simple absence reporting.
4	Development team	The four team members who analyze, design, implement, and test the system. Not an application user.	Clear, feasible, and testable requirements; stable scope; manageable technical risk (scheduling constraints, data consistency).
5	Lecturer / TA	Evaluates the project documents, milestones, and demonstration. Not an application user.	Complete and consistent documentation; requirements aligned with the four Must-have features (F01–F04); a working prototype.

5 Requirements

5.1 Functional Requirements

ID	Requirement	Feature	Priority	Source
FR-01	The system shall authenticate users via Email and Password using Supabase Auth.	F01	Must	PA1 Vision / UC-01
FR-02	The system shall enforce role-based authorization (Employee, Manager, Boss) on both the front-end user interface and back-end REST API endpoints.	F01	Must	PA1 Vision / UC-01
FR-03	The system shall end the application session on logout, clear local session credentials, and reject expired tokens on protected requests.	F01	Must	UC-01
FR-04	The system shall generate a weekly draft that respects all applicable hard constraints and prioritizes staffing coverage. It shall report any uncovered positions against the staffing requirements.	F02	Must	PA1 Scope / UC-05
FR-05	The system shall evaluate workload fairness over a five-week rolling window consisting of the target scheduling week and up to four preceding official weeks. It shall evaluate scheduled working-hour deviation, night-shift imbalance, and Prefer Not conflicts. Missing historical weeks shall be identified explicitly and shall not be treated as zero-workload observations.	F02	Must	PA1 Vision / UC-05
FR-06	The system shall allow the Manager to manually adjust draft shift assignments while automatically re-validating hard constraints.	F02	Must	UC-05
FR-07	The system shall identify and display uncovered staffing positions, requiring explicit Manager confirmation prior to publishing a partially covered schedule.	F02	Must	UC-05
FR-08	The system shall maintain an immutable version history of published schedules and allow planned revisions to future shifts without destroying past historical versions.	F02	Must	PA1 Scope / UC-05
FR-09	The system shall display published official rosters based on user role: individual shifts for	F03	Must	PA1 Scope / UC-06

	Employees, team roster for Managers, and read-only overview for Boss users.			
FR-10	The system shall identify new, changed, and cancelled assignments between consecutive published schedule versions and display, for each affected Employee, the cumulative changes since their last acknowledged applicable version, while preserving access to intermediate version history.	F03	Must	UC-06
FR-11	The system shall allow an affected Employee to explicitly acknowledge the latest applicable published schedule update after reviewing the changes. Acknowledgement shall be associated with the Employee and the applicable schedule version; viewing a schedule alone shall not count as acknowledgement.	F03	Must	PA1 Scope / UC-07
FR-12	The system shall display acknowledgement progress to authorized Managers and Boss users, distinguishing Acknowledged, Pending, Superseded, and Not Required outcomes as applicable. The required acknowledgement count shall include only Employees affected by the corresponding published update.	F03	Must	UC-06, UC-07
FR-13	The system shall allow Employees to report an absence for their own assigned future or ongoing shift that has not ended, providing a mandatory reason category and optional notes. The system shall record the report and create or reuse the related coverage request without automatically assigning a replacement.	F04	Must	PA1 Scope / UC-08
FR-14	The system shall allow an authorized Manager to record a no-show after an assigned shift starts and create or reuse its unresolved coverage request under BR-08.	F04	Must	UC-09
FR-15	The system shall show only candidates who satisfy the applicable hard constraints in BR-03 and BR-07 for the proposed replacement interval. The absent Employee is excluded from their own coverage request.	F04	Must	PA1 Scope / UC-09
FR-16	The system shall allow the Manager to issue a replacement offer to an eligible Employee while enforcing a maximum of one active Pending or Accepted offer per uncovered staffing position.	F04	Must	UC-09

FR-17	The system shall allow the target candidate to Accept or Decline a received replacement offer.	F04	Must	PA1 Scope / UC-09
FR-18	The system shall allow an authorized Manager to confirm an accepted offer after revalidation. It shall atomically update the effective assignment, offer, coverage request, current official schedule version, and required acknowledgements. For an ongoing shift, the replacement applies only from confirmation through the remaining valid interval.	F04	Must	PA1 Scope / UC-09
FR-19	The system shall allow an authorized Manager to create, update, activate, and deactivate Employee accounts within their assigned production team. Accounts created through this workflow shall receive the Employee role; unauthorized role elevation and public self-registration shall not be permitted.	F01	Must	UC-02; Account management policy
FR-20	The system shall allow Employees to submit and update their own Available, Unavailable, and Prefer Not declarations for future scheduling weeks before both the configured cutoff and the start of draft generation. When no weekly declaration exists, the applicable baseline availability shall be used.	F02	Must	UC-03; BR-02
FR-21	The system shall allow an authorized Manager to create and maintain reusable staffing requirement templates and week-specific configurations specifying non-negative required headcounts by shift and position. Saving a configuration shall not retroactively alter a published schedule.	F02	Must	UC-04; Staffing configuration policy
FR-22	The system shall allow an authorized Manager to post overtime shifts, including holiday shifts, and approve or reject voluntary Employee sign-ups. It shall record each Employee's consent to the specific offered terms and allow withdrawal before assignment. The system shall validate consent, approval and eligibility under BR-07 before generating, editing, publishing or confirming an overtime assignment. For overtime replacement, the candidate's explicit consent to the offer and the Manager's final confirmation satisfy the sign-up and approval steps.	F02	Must	Project owner demo direction, BR-07, UC-03/04/05/09

5.2 Non-Functional Requirements

ID	Category	Requirement (measurable)	How it will be verified
NFR-01	Performance	P95 navigation-to-usable-content time for login, My Schedule, Team Schedule, Availability, and Staffing Requirements shall be at most 3 s. P95 response time for availability, staffing, absence, acknowledgement, and offer-response API requests shall be at most 1 s. Draft generation shall finish within 30 s, and a generation failure shall preserve the existing official data.	Measure at least 100 requests per operation for the page and API targets. Run generation on 10 documented sample datasets; every run must finish within 30 s.
NFR-02	Security	The deployed client and API shall use HTTPS. Application tables and logs shall not contain plaintext passwords or session credentials; credentials are handled by Supabase Auth.	Inspect the application tables, the logging configuration, and sampled logs. Verify the deployed HTTPS endpoints.
NFR-03	Accessibility	Every input has a visible associated label. Text contrast is at least 4.5:1. Main flows work with the keyboard, focus remains visible, and status is not conveyed by colour alone.	Run each main flow with the keyboard only, inspect labels and contrast, and inspect a 375 px viewport. A Lighthouse score of at least 90 is a supporting target, not a substitute for these checks.
NFR-04	Authorization	Each protected API enforces authentication, role, and object ownership. Unauthenticated or invalid-token requests return 401; authenticated forbidden operations return 403 without restricted data. Boss access is read-only and excludes personal absence explanations.	Call the APIs directly as all three roles. Attempt to read another Employee's roster, availability, absence details, and offer response.
NFR-05	Consistency	Repeated or concurrent submission of the same logical operation shall not create duplicate acknowledgements, reports, coverage requests, or confirmed replacements. A failure during confirmation shall leave the	Send at least two simultaneous identical requests and competing confirmations. Inject a failure during confirmation and check that all four items are unchanged.

		assignment, the current schedule version, the offer outcome, and the coverage status unchanged. At most one published version is current per team and week.	

5.3 Domain Requirements and Business Rules

ID	Rule	Source (policy, law, standard)
BR-01	**Role Authorization & Access Control:** System functionality is strictly partitioned across three roles: Employee (personal roster view, availability submission, absence reporting, replacement offer response), Manager (staffing configuration, draft schedule generation/adjustment, schedule publication, replacement confirmation), and Boss (read-only team-wide coverage and workload audit view). Administrative and schedule editing privileges must be enforced on the server for every REST API request using verified JWT role claims. An authorized Manager may create and manage Employee accounts within the assigned team but may not create privileged Manager or Boss accounts or elevate an Employee to either privileged role. Initial privileged accounts are provisioned separately.	· ISO/IEC 27002:2022, Control 8.2 (Privileged access rights). · NIST SP 800-53 Rev. 5, Control AC-3 (Access Enforcement).
BR-02	**Employee Availability and Submission Cutoff:** Employees may create or update availability declarations for future scheduling weeks before the configured Thursday 23:59 cutoff and before schedule generation begins. The server shall enforce the cutoff when a submission is received. Available, Unavailable, and Prefer Not are distinct declaration statuses. An Employee may declare all shifts Unavailable; the system shall warn about potential staffing shortages but shall not silently override the declaration. Weeks without submitted declarations shall use the applicable baseline availability profile.	· ShiftManager draft weekly availability policy (FR-20, UC-03).
BR-03	**Hard and Soft Scheduling Constraints:** All generated assignments, manual draft changes, and confirmed replacements must satisfy the applicable hard constraints: no overlapping assignments; required qualifications; effective Unavailable periods; required rest between shifts, including across week boundaries; and applicable working-time and overtime conditions. The scheduling policy shall enforce at least 12 hours	· Vietnam Labour Code No. 45/2019/QH14, Articles 109 and 110 (breaks and rest between shifts). See BR-07 for working-time and overtime rules.

	between consecutive assigned shifts. Prefer Not is a soft preference and may be overridden with an explicit optimization penalty, but no hard constraint may be relaxed to increase staffing coverage. If mandatory eligibility or compliance information is unavailable, the assignment shall not be treated as validated.	• ISO 45001:2018 Occupational Health & Safety Management – https://www.iso.org/standard/63787.html
BR-04	**Five-Week Workload Fairness:** The scheduling fairness window consists of the target week and up to four preceding official weeks. Workload evaluation shall consider scheduled working hours relative to each Employee's applicable target, distribution of night shifts among comparable eligible Employees, and assignments conflicting with Prefer Not preferences. The system shall identify the number of historical weeks actually available. Missing historical weeks shall not be inserted as zero-hour observations. Staffing coverage shall take priority over soft fairness and preference objectives; hard constraints shall never be relaxed.	• ShiftManager draft workload-balancing policy (FR-05, UC-05).
BR-05	**Shortage Schedule Publication & Warning:** A weekly draft schedule containing uncovered staffing positions (shortages) may only be published if the Manager receives an explicit system warning and issues a formal override confirmation. Uncovered positions must remain visibly highlighted as "Uncovered" across Manager and Boss overview dashboards until fully resolved.	• ShiftManager draft shortage-publication policy (FR-07, UC-05).
BR-06	**Replacement Coverage and Offer Lifecycle:** A replacement requires a Manager-issued offer, acceptance by the addressed Employee, and final Manager confirmation after eligibility revalidation. Offer states are Pending, Accepted, Declined, Cancelled, Expired, and Confirmed. No more than one Pending or Accepted offer may exist for the same uncovered position. The coverage request has a separate lifecycle: Unresolved, Resolved, Cancelled, or Unresolved at Shift End. Confirming a replacement must atomically update the effective assignment, offer, coverage request, official schedule version, and required acknowledgement updates. Previously effective assignment intervals must remain traceable. A request that still needs coverage at shift end is recorded as Unresolved at Shift End.	• ShiftManager draft coverage and offer policy (FR-14–FR-18, UC-09). • References: ISO/IEC 9075-2:2023 — SQL/Foundation. For atomicity and rollback, see PostgreSQL documentation — Transactions.
BR-07	**Working Time, Rest, and Overtime Eligibility:** The system classifies ordinary time and overtime using the configured work schedule and applicable rules. The	Vietnam Labour Code No. 45/2019/QH14, Articles

	Manager posts the date, time, location, duties and any informational compensation note. The Employee may opt in, and the system records consent, timestamp and offer terms. The Manager approves or rejects the sign-up. The Employee may withdraw before assignment. A material change to offer terms requires renewed consent and approval. The demo may display a Manager-entered note such as 300% for a Tết shift. Payroll remains outside product scope. Overtime assignments may be approved only when the applicable employee consent, working-time limits, overtime limits and other statutory conditions can be verified using relevant records. Availability declarations do not constitute overtime consent. Rest and working-time checks shall consider adjacent-week assignments and any relevant monthly or annual history. The five-week fairness history is not sufficient evidence of legal overtime compliance. The system checks recorded consent, actual prior work and overtime totals, scheduled commitments, the proposed interval, and applicable rest and working-time limits. Demo input records supply actual-hour totals. Missing required data, invalid consent or a failed limit blocks assignment and exposes the reason.	105, 107, 110–112 (consolidated text 18/VBHN-VPQH). Decree No. 145/2020/ND-CP, Articles 59–60, covers consent and overtime limits. The sign-up and approval steps are ShiftManager demo policies. The 300% note in Figure 8.9 is illustrative. See Article 98(1)(c) of the Labour Code for holiday overtime pay.
BR-08	**Absence Reporting and No-Show Integrity:** An Employee may report an absence for their own assigned shift that has not ended. A reason category is mandatory; explanatory notes are optional. Reports for an ongoing shift must preserve previously effective assignment history. A Manager may record a no-show only after the shift has started. Duplicate reports or no-show records for the same unresolved assignment must not create duplicate coverage requests. Every report and Manager-recorded no-show shall remain traceable to its actor and timestamp.	Proposed ShiftManager absence-reporting policy; FR-13–FR-14; UC-08/09
BR-09	**Schedule Acknowledgement Applicability:** A new official schedule version requires acknowledgement only from Employees whose applicable assignments are added, changed, or cancelled. On first publication, all published assignments count as new. Affected Employees are Pending until they explicitly acknowledge the latest applicable version; unaffected Employees are Not Required for that update. Older pending applicable versions become Superseded when replaced by a newer applicable update. Historical	Proposed ShiftManager acknowledgement policy; FR-10–FR-12; UC-06/07

	version changes remain accessible, while the latest acknowledgement screen may summarize cumulative changes since the Employee's last acknowledged applicable version. The acknowledgement denominator counts only Employees required to acknowledge that update.	

5.4 Acceptance Criteria

The acceptance criteria below are derived from the current functional requirements (FR-01–FR-22), business rules, and use case specifications (UC-01–UC-09). AC-01–AC-23 cover the initial requirements by feature. AC-24–AC-43 extend coverage for account management, availability, staffing configuration, and cross-feature rules. AC-44–AC-49 cover voluntary overtime. All are proposed scenarios for PA4. Section 10 records the review outcome and implementation follow-up.

F01 – Authentication and Role-Based Access

AC-01 (FR-01; UC-01)

- **Given** an active pre-created user account with valid credentials
- **When** the user submits the login form
- **Then** the system authenticates the user and opens the interface authorized for that user.

AC-02 (FR-01; UC-01)

- **Given** an incorrect password or a non-existing account
- **When** the user attempts to log in
- **Then** no authenticated session is created and a non-disclosing error message is shown.

AC-03 (FR-02; UC-05)

- **Given** an authenticated Employee without schedule management permission
- **When** the Employee attempts to generate or publish a team schedule

- **Then** the system rejects the action on the server and does not disclose restricted scheduling data.

AC-04 (FR-03; UC-01)

- **Given** an authenticated user with an active application session
- **When** the user selects Log Out
- **Then** the application ends the local session, clears its local credentials, and requires valid authentication for further protected access. The server rejects requests with expired credentials.

F02 – Constraint-Based Automatic Scheduling

AC-05 (FR-04; UC-03/05)

- **Given** an Employee has effective Unavailable and Prefer Not entries for a target week
- **When** the system generates a schedule draft
- **Then** it excludes effective Unavailable periods as hard constraints and treats Prefer Not periods as soft preferences.

AC-06 (FR-04; UC-04/05)

- **Given** a staffing template with shift- and position-specific staffing requirements
- **When** the Manager generates a schedule for a target week
- **Then** the draft reports assignments and any uncovered positions against the configured requirements.

AC-07 (FR-04; UC-05)

- **Given** valid employees, qualifications, availability, and staffing requirements
- **When** the Manager requests automatic scheduling
- **Then** the system returns a draft without overlapping or otherwise hard-constraint-violating employee assignments.

AC-08 (FR-05; UC-05; BR-04)

- **Given** a target scheduling week and between zero and four available preceding official weeks
- **When** the system evaluates a generated draft
- **Then** it displays the applicable workload, night-shift, and Prefer Not indicators together with the number of historical weeks used, without counting unavailable historical weeks as zero-workload observations.

AC-09 (FR-06; UC-05)

- **Given** a valid draft and an eligible substitute for an assignment
- **When** the Manager makes a manual draft adjustment
- **Then** the adjustment is accepted and the coverage and fairness indicators are recalculated.

AC-10 (FR-06; UC-05)

- **Given** a draft and a manual assignment that conflicts with an effective Unavailable period
- **When** the Manager attempts to save the change
- **Then** the system rejects that assignment, states the constraint violated and retains the previous draft.

AC-11 (FR-07; UC-05)

- **Given** a draft with at least one uncovered position
- **When** the Manager requests publication
- **Then** the system shows the shortage and requires explicit confirmation before publication proceeds.

AC-12 (FR-07; UC-05)

- **Given** the shortage confirmation is displayed

- **When** the Manager declines to publish a partially covered draft
- **Then** the draft remains unpublished and any currently official schedule remains unchanged.

AC-13 (FR-08; UC-05)

- **Given** a published weekly schedule
- **When** the Manager publishes an approved planned revision affecting future shifts
- **Then** the new version becomes current, only one official version is current and the prior version remains accessible in history.

F03 – Published Schedule and Acknowledgement

AC-14 (FR-09; UC-06)

- **Given** authenticated Employee, Manager and Boss users
- **When** each opens their schedule or overview screen
- **Then** each sees only the personal, team or read-only summary data allowed by their role.

AC-15 (FR-10; UC-06)

- **Given** consecutive published schedule versions and an affected Employee with an earlier acknowledged version
- **When** the Employee views the latest applicable update
- **Then** the system distinguishes new, changed, and cancelled assignments, presents cumulative changes since the last acknowledged applicable version, and retains intermediate version history.

AC-16 (FR-11; UC-07)

- **Given** an Employee has a Pending acknowledgement for the latest applicable schedule version
- **When** the Employee reviews the update and explicitly selects Acknowledge

- **Then** one acknowledgement is recorded for that Employee and applicable version, and merely viewing the schedule does not count as acknowledgement.

AC-17 (FR-12; UC-06/07)

- **Given** a published update affects only some Employees in a team
- **When** an authorized Manager or Boss views acknowledgement progress
- **Then** the required-count denominator includes only affected Employees. The view distinguishes Acknowledged, Pending, Superseded, and Not Required wherever applicable.

F04 – Absence Reporting and Shift Coverage

AC-18 (FR-13; UC-08)

- **Given** an Employee owns an assigned future or ongoing shift that has not ended
- **When** the Employee submits an absence report with a reason category and optional notes
- **Then** the system records the report, creates or reuses the associated unresolved coverage request, and does not automatically assign a replacement.

AC-19 (FR-14; UC-09)

- **Given** a scheduled shift has started and its assigned Employee has not appeared
- **When** the Manager records the no-show
- **Then** the system stores the observation and identifies the affected coverage position.

AC-20 (FR-15; UC-09)

- **Given** an uncovered position and a list of candidate Employees
- **When** the Manager requests replacement suggestions
- **Then** only candidates meeting all applicable hard constraints in BR-03 and BR-07 are shown as eligible.

AC-21 (FR-16; UC-09)

- **Given** an uncovered staffing position already has a Pending or Accepted replacement offer
- **When** the Manager attempts to issue another offer for the same position
- **Then** the system rejects the second offer and preserves the existing active offer.

AC-22 (FR-17; UC-09)

- **Given** a candidate has received a Pending replacement offer
- **When** the candidate selects Accept or Decline
- **Then** the corresponding response is recorded; an unanswered or declined offer does not create an official assignment.

AC-23 (FR-18; UC-09)

- **Given** an accepted replacement offer and a currently eligible candidate
- **When** the Manager confirms the replacement
- **Then** the system commits one consistent assignment change, offer confirmation, coverage outcome, schedule version, and required acknowledgement updates; repeated or competing confirmations do not create duplicate assignments, and a failed transaction leaves no partial committed changes.

AC-24 (FR-19; UC-02)

- **Given** an authorized Manager and valid information for a new Employee
- **When** the Manager creates an Employee account
- **Then** the system provisions an Employee identity and profile for the correct team, assigns the Employee role, and records the operation without creating a privileged account.

AC-25 (FR-19; UC-02)

- **Given** a Manager without authority to administer privileged roles

- **When** the Manager attempts to create a Boss or Manager account, elevate an Employee's role, or modify an account outside the assigned team
- **Then** the server rejects the operation and does not make an unauthorized account change.

AC-26 (FR-20; UC-03)

- **Given** an eligible future scheduling week before its submission cutoff and before draft generation
- **When** the Employee saves valid availability declarations
- **Then** the declarations are saved and used as scheduling inputs for that week.

AC-27 (FR-20; UC-03; BR-02)

- **Given** an Employee opened the availability form before the cutoff
- **When** the submission reaches the server after the cutoff or after scheduling has been locked
- **Then** the server rejects the submission and preserves the previously saved declaration.

AC-28 (FR-20; UC-03; BR-02/03)

- **Given** a scheduling week is still open for declarations
- **When** an Employee marks every shift as Unavailable and confirms the warning
- **Then** the declaration is saved and automatic scheduling does not assign that Employee during those Unavailable periods.

AC-29 (FR-21; UC-04)

- **Given** valid staffing requirements by day, shift, and position
- **When** the Manager saves a default or week-specific configuration
- **Then** the selected configuration is persisted without automatically modifying any published schedule.

AC-30 (FR-21; UC-04)

- **Given** a staffing configuration contains a negative value, non-integer value, or value exceeding an approved configured limit
- **When** the Manager attempts to save the configuration
- **Then** the system identifies invalid fields and does not persist an invalid configuration. A zero value is valid where the applicable operating policy permits it.

AC-31 (FR-04/06/15/18; BR-07)

- **Given** a proposed assignment would violate an applicable working-time limit, required rest period, or overtime eligibility condition, including an assignment across a week boundary
- **When** the assignment is evaluated during generation, manual editing, or replacement confirmation
- **Then** the system rejects the invalid assignment, displays the applicable reason, and retains any resulting staffing shortage.

AC-32 (FR-13; UC-08)

- **Given** an Employee is assigned an ongoing shift that has not ended
- **When** the Employee reports an absence with a valid reason category
- **Then** the system records an urgent coverage request without retroactively changing previously effective assignment intervals.

AC-33 (FR-13; BR-08)

- **Given** an absence report does not contain a required reason category
- **When** the Employee submits the form
- **Then** the system rejects the submission and does not create a new absence record or coverage request. Optional notes are not mandatory.

AC-34 (FR-13/14; BR-08)

- **Given** an unresolved coverage request already exists for an assignment
- **When** a duplicate absence report or no-show event is submitted for that same unresolved assignment
- **Then** the system reuses the existing coverage request rather than creating a duplicate.

AC-35 (FR-16/17; UC-09)

- **Given** an offer for an uncovered position is Accepted but not yet Confirmed
- **When** the Manager attempts to issue another replacement offer for the same position
- **Then** the system blocks the new offer until the existing active offer is cancelled, expired, or otherwise legitimately concluded.

AC-36 (FR-17/18; UC-09)

- **Given** an offer is Cancelled, Expired, or addressed to a different Employee
- **When** acceptance or confirmation is attempted by an unauthorized or ineligible actor
- **Then** the system rejects the action and creates no replacement assignment.

AC-37 (FR-18; NFR-05; UC-09)

- **Given** two simultaneous confirmation attempts or an injected transaction failure
- **When** replacement confirmation is processed
- **Then** at most one consistent replacement is committed, and any failed transaction rolls back all related changes.

AC-38 (FR-11/12; BR-09)

- **Given** a new official team schedule version does not affect an Employee's assignments
- **When** that version is published
- **Then** the Employee does not receive a new required acknowledgement, and the required-count denominator does not increase because of that Employee.

AC-39 (FR-10/11; UC-06/07)

- **Given** several applicable schedule changes have occurred since an Employee's last acknowledgement
- **When** the Employee reviews the latest applicable update
- **Then** cumulative changes are available, intermediate version history remains accessible, and earlier unacknowledged applicable versions are Superseded.

AC-40 (FR-11/12; UC-07; BR-09)

- **Given** the first official schedule publication contains assignments for some Employees but none for others
- **When** publication is completed
- **Then** affected Employees are Pending acknowledgement and Employees with no applicable update are Not Required.

AC-41 (FR-18; UC-09; BR-06)

- **Given** an affected shift runs from 06:00 to 14:00 and a valid replacement is confirmed at 10:00
- **When** the replacement transaction is committed
- **Then** the substitute's effective assignment covers only 10:00–14:00 and the earlier interval remains in historical records.

AC-42 (FR-13/14/18; UC-06/09)

- **Given** an absence or no-show has left an assigned staffing interval uncovered
- **When** an authorized Manager or Boss views staffing coverage
- **Then** the uncovered interval remains visible even if the original assignment record has been retained for traceability.

AC-43 (FR-05; BR-04)

- **Given** a target scheduling week has incomplete historical workload information
- **When** the system compares and displays valid draft schedules
- **Then** it reports the actual historical weeks used, scheduled-hour deviation, night-shift distribution, and Prefer Not conflicts without treating missing historical records as zero-hour observations.

Voluntary Overtime and Overtime Replacement

AC-44 (FR-22, UC-03/04/05)

- **Given** a Manager posts a holiday overtime shift with its time, duties, location and an informational compensation note
- **When** one Employee explicitly signs up and another takes no action
- **Then** the system records consent for the first Employee and the offered terms. The second Employee has no sign-up or consent.

AC-45 (FR-22, UC-05, BR-07)

- **Given** a valid Employee sign-up and complete eligibility records
- **When** the Manager approves it and requests generation or manual assignment
- **Then** the Employee may be considered for that shift if all applicable constraints pass.

AC-46 (FR-22, UC-03/05, BR-07)

- **Given** a sign-up is pending, rejected, withdrawn, or tied to superseded offer terms
- **When** the system evaluates the Employee for overtime
- **Then** it blocks assignment and explains the missing approval or consent.

AC-47 (FR-22, UC-05, BR-07)

- **Given** an approved sign-up and missing hour records or a projected working-time, overtime or rest violation

- **When** the system validates the proposed assignment
- **Then** it blocks the assignment, shows the reason and preserves any staffing shortage.

AC-48 (FR-18/22, UC-09, BR-07)

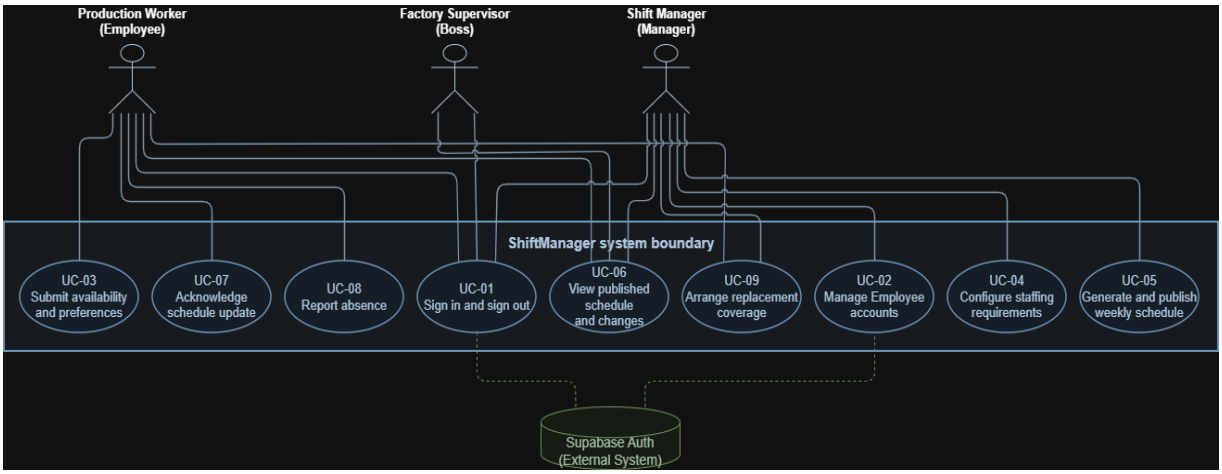
- **Given** an ongoing shift requires an overtime replacement and the candidate explicitly consents to the applicable offer terms
- **When** the Manager confirms the accepted offer after eligibility revalidation
- **Then** the system commits the replacement for the confirmed remaining interval and preserves earlier assignment history.

AC-49 (FR-18/22, UC-09, BR-07)

- **Given** a proposed overtime replacement lacks valid consent or the candidate becomes ineligible
- **When** the Manager attempts confirmation
- **Then** the system rejects confirmation and preserves the existing official schedule and coverage state.

6 Use Case Model

6.1 Use Case Diagram



6.2 Actors

Actor	Description
Production Worker (Employee)	Factory production line worker who logs in, submits availability/preferences, views personal rosters, acknowledges schedule updates, reports shift absences, and responds to replacement offers.
Shift Manager (Manager)	Production team supervisor who configures staffing demand, triggers automated roster generation, reviews and publishes schedules, records no-shows, and confirms replacement coverage.
Supervisor (Boss)	Plant management executive who accesses read-only dashboards to monitor team-wide staffing coverage, open shortages, absence summaries, and workload distribution.
Supabase Auth	External identity authentication provider handling credential verification, JWT session token generation, and account status management.

6.3 Use Case List

ID	Use case	Primary actor	Priority	Related FR
UC-01	Log In to ShiftManager	Employee, Manager, Boss	Must	FR-01, FR-02, FR-03

UC-02	Manage User Accounts and Roles	Manager	Must	FR-01, FR-02, FR-19
UC-03	Submit Availability and Preferences	Employee	Must	FR-04, FR-05, FR-20 , FR-22
UC-04	Configure Staffing Requirements	Manager	Must	FR-04, FR-21 , FR-22
UC-05	Generate and Publish Weekly Schedule	Manager	Must	FR-04, FR-05, FR-06, FR-07, FR-08 , FR-22
UC-06	View Published Schedule and Changes	Employee, Manager, Boss	Must	FR-09, FR-10, FR-12 (F03); BR-09 applies to acknowledgement visibility.
UC-07	Acknowledge Schedule Update	Employee	Must	FR-10, FR-11, FR-12
UC-08	Report Absence	Employee	Must	FR-13
UC-09	Arrange Replacement Coverage	Manager	Must	FR-14, FR-15, FR-16, FR-17, FR-18 , FR-22

7 Use Case Specifications

7.1 Use Case UC-01

Use case ID	UC-01
Use case name	Log In to ShiftManager
Brief description	A pre-registered user (Employee, Manager, or Boss) authenticates with an email address and a password. After successful authentication, the system establishes a session, identifies the user's role, and presents only the functions and data permitted for that role. The use case also covers logging out and denying access to functions outside the user's role.
Primary actor	Employee, Manager, or Boss (any registered user)
Secondary actors	Authentication service (Supabase Auth)
Priority	Must
Related requirements	FR-01, FR-02, FR-03 (F01 – Authentication & Role-Based Access).
Trigger	A user opens ShiftManager without an active session, or selects Log In.
Pre-conditions	The user has a pre-created, active account with exactly one assigned role (Employee, Manager, or Boss). The application and the authentication service are reachable.
Post-conditions	**Success**: 1. An authenticated session exists for the user. 2. The user is shown the home page of their role: My Schedule (Employee), Weekly Scheduling (Manager), or Staffing Overview (Boss). 3. Every subsequent request is authorized according to the user's role. **Failure or Cancellation:** 1. No session is created. 2. No schedule, employee, or role-specific data is disclosed to the unauthenticated user. 3. The user remains on the login page.
Main flow	1. The user opens the ShiftManager login page. 2. The system displays the login form with Email and Password fields. 3. The user enters the email address and password and selects Log In. 4. The system checks that both fields are filled in and that the email address has a valid format. 5. The system sends the credentials to the authentication service for verification. 6. The authentication service confirms the credentials and returns a session token. 7. The system retrieves the user's profile and assigned role. 8. The system establishes the session and redirects the user to the home page of the role.

Alternative flows	**A1 – Valid Session Already Exists (at Step 1)** 1. The user opens the application with a valid, unexpired session. 2. The system skips the login form and continues at Step 7. **A2 – Return to the Requested Page After Session Expiry (at Step 8)** 1. The user was redirected to the login page because a previous session expired while requesting a specific page. 2. After successful authentication, the system returns the user to the requested page if the user's role permits it; otherwise it shows the home page of the role. **A3 – Log Out (after successful authentication)** 1. The user selects Log Out. 2. The application signs out the current session, clears locally stored session credentials, removes protected content, and prevents the session from being refreshed. 3. The system returns to the login page and confirms that the user has logged out. The MVP does not promise immediate revocation of previously issued access tokens on every device; such tokens remain bounded by their configured expiry.
Exception flows	**E1 – Missing or Malformed Input (at Step 4):** The email address or password is empty, or the email format is invalid. The system highlights the field, explains what must be corrected, and does not send the credentials for verification. **E2 – Invalid Credentials (at Step 6):** The authentication service rejects the credentials. The system shows a generic message ("Invalid email or password") that does not reveal which part was wrong, clears the password field, and creates no session. **E3 – Account Disabled or No Role Assigned (at Step 7):** The credentials are valid, but the account is inactive or has no role. The system denies access, tells the user to contact the Manager, and does not establish a session. **E4 – Authentication Rate Limit (at Step 6):** The authentication service limits repeated attempts. The system shows the retry message returned by the service and creates no session. The use case does not define its own lockout duration. **E5 – Authentication Service Unavailable (at Steps 5–6):** The service cannot be reached or does not respond. The system reports that login is temporarily unavailable, creates no session, and lets the user try again. **E6 – Access Beyond Role (after successful authentication):** The user requests a page or action that the role does not permit (for example, an Employee opens Weekly Scheduling). The system denies the request, shows a "no permission" message, discloses no restricted data, and keeps the session active.
Business rules	1. Initial privileged demonstration accounts are provisioned before system use. Additional Employee accounts may be created by an authorized Manager through UC-02. Public self-registration is not supported. 2. Each user has exactly one role: Employee, Manager, or Boss.

	3. An Employee may view their own roster and relevant schedule changes; a Manager may manage the schedules of their own production team; a Boss may access authorized staffing and workload summaries. 4. Role permissions are enforced on the server for every request, not only by hiding menu items. 5. A failed-login message must not reveal whether an email address is registered.
Non-functional constraints	• Passwords are handled only by Supabase Auth; the deployed client and API use HTTPS (NFR-02). • Every protected API request shall validate the token signature, expiry, and authorized role (NFR-04). When no valid session can be obtained, the application returns the user to login. No custom inactivity timeout or account lockout duration is specified in the MVP. • The login page shall load within the target in NFR-01 and meet the accessibility checks in NFR-03 (labelled fields, keyboard use, 375 px layout).

7.2 Use Case UC-02

Use case ID	UC-02
Use case name	Manage User Accounts and Roles
Brief description	The Manager manages Employee accounts belonging to the assigned production team. The Manager can create accounts, update permitted profile information, activate or deactivate Employee access, and view the account status. Initial Manager and Boss accounts are provisioned separately. This use case does not permit public registration or privilege escalation.
Primary actor	Manager
Secondary actors	Supabase Auth (Authentication Service)
Priority	Must
Related requirements	FR-01, FR-02, FR-19 (F01)
Trigger	The Manager opens Employee Account Management and selects an account management action.
Pre-conditions	- The Manager is authenticated and authorized for the assigned production team. - Supabase Auth and the application database are available. - For account modification, the target Employee account belongs to the Manager's authorized team.
Post-conditions	**Success:** 1. The requested account change is completed and its outcome is confirmed. 2. The Employee's authentication identity, application profile, role, and active status are consistent. 3. The operation is recorded with the responsible Manager and timestamp. 4. Existing scheduling and activity history is preserved. **Failure or Cancellation:**

	1. No unauthorized account change takes effect. 2. The system does not report incomplete provisioning as successful. 3. Existing valid account data remains unchanged or a safe recovery action is recorded.
Main flow	1. The Manager opens Employee Account Management. 2. The system displays Employee accounts within the Manager's team, including their account status. 3. The Manager selects Create Employee Account. 4. The system displays the required account and profile fields. 5. The Manager enters a name, email address, applicable team information, and work qualifications. 6. The Manager submits the creation request. 7. The system validates the inputs, verifies team authorization, and checks whether the account already exists. 8. The system securely provisions an Employee authentication identity and an associated application profile. 9. The system assigns the Employee role and links the profile to the Manager's team. 10. The system records the operation and confirms successful account creation.
Alternative flows	**A1 – Update Employee Profile (at Step 3):** 1. The Manager selects an existing Employee account. 2. The system displays the permitted editable fields. 3. The Manager updates the information. 4. The system validates and saves the changes. 5. The system records and confirms the update. **A2 – Deactivate Employee Account (at Step 3):** 1. The Manager selects an active Employee account. 2. The system displays a confirmation prompt. 3. The Manager confirms deactivation. 4. The system marks the account inactive and prevents further authorized application access. 5. The system preserves historical records and logs the operation. **A3 – Reactivate Employee Account (at Step 3):** 1. The Manager selects an inactive Employee account. 2. The Manager confirms reactivation. 3. The system validates authorization and reactivates access. 4. The system records the action and confirms the result. **A4 – Cancel Operation (at Step 5 or Step 6):** 1. The Manager cancels the action. 2. The system discards unsaved changes. 3. No account modification is committed.

Exception flows	**E1 – Duplicate Email (at Step 7):** An account with the same email already exists. The system rejects duplicate creation and displays an appropriate message. **E2 – Invalid Input (at Step 7):** Required information is missing or invalid. The system identifies the affected fields and prevents submission until corrected. **E3 – Unauthorized Account or Role Operation (at Step 7):** The Manager attempts to change an account outside the authorized team, create a privileged Manager/Boss account, or elevate an Employee's role. The system denies the operation without disclosing restricted data. **E4 – Provisioning Failure (at Steps 8–9):** Authentication or profile creation fails. The system prevents an incomplete identity/profile from being reported as an active account and records the failure for safe recovery. **E5 – Concurrent Account Change:** The target account has been changed since the page was loaded. The system prevents an outdated update from silently overwriting newer information. **E6 – Session Expired:** The Manager is no longer authenticated. The system rejects the protected operation and requires login.
Business rules	1. Only an authorized Manager may manage Employee accounts belonging to the assigned team. 2. Accounts created through this use case receive the Employee role. 3. The Manager cannot create Manager/Boss accounts or grant elevated privileges. 4. Public self-registration is disabled in the MVP. 5. Deactivation prevents access without deleting historical schedules, acknowledgement records, or absence records. 6. Account management operations must record the actor, affected account, action, and timestamp. 7. Existing scheduling constraints must remain enforceable when an account is deactivated or its qualifications change.
Non-functional constraints	1. All privileged operations require server-side role and team authorization (NFR-04). 2. Credentials shall be handled by the authentication service; plaintext passwords shall not be stored in application tables (NFR-02). 3. Account forms must have accessible labels, keyboard support, and clear error messages (NFR-03). 4. Concurrent or failed provisioning operations must not leave an incorrectly active account. 5. Account-management actions must be auditable.

7.3 Use Case UC-03

Use case ID	UC-03
Use case name	Submit Availability and Preferences
Brief description	An Employee submits or updates their weekly shift availability (Available / Unavailable) and preferences for upcoming scheduling periods. This data feeds into the constraint solver for automated draft generation (UC-05).

Primary actor	Production Worker (Employee)
Secondary actors	None
Priority	Must
Related requirements	FR-04, FR-05, FR-20 and FR-22. When a Manager has posted an overtime shift, the Employee can open the offer, review its terms, and explicitly sign up or withdraw before assignment. The system records the response and shows the approval status separately from weekly availability.
Trigger	An Employee accesses the availability settings page prior to the schedule generation cutoff date.
Pre-conditions	1. The Employee is authenticated into ShiftManager (UC-01). 2. The target week is open for availability declarations (draft generation has not started).
Post-conditions	**Success:** 1. Employee availability and preferences for the target week are saved to the database. 2. Updated availability is immediately accessible to the schedule generation engine (UC-05). 3. The submission timestamp is recorded. **Failure / Cancellation:** 1. No changes are saved to the database. 2. Existing baseline availability rules remain
Main flow	1. Employee opens the Availability & Preferences page. 2. System displays default baseline availability and current declaration status for upcoming weeks. 3. Employee selects a target scheduling week. 4. System displays a 7-day x 3-shift matrix (Morning, Afternoon, Night). 5. Employee toggles slot statuses ("Available", "Unavailable", or "Prefer Not"). 6. Employee selects Submit Availability. 7. The server verifies ownership, slot values, the submission cutoff, and that draft generation has yet to begin. 8. System stores the submission record and updates declaration status to "Submitted". 9. System displays a confirmation message with the update timestamp.
Alternative flows	**A1 - Revert to Default Baseline (at Step 5)**: 1. Employee chooses to reset declarations to corporate baseline rules.

	2. If the week is still open, the system clears the custom declaration and uses baseline availability. **A2 - Update Prior to Cutoff (at Step 3)**: 1. Employee re-opens an already submitted week before scheduling lock-in. 2. Employee modifies slot choices and re-submits. 3. System overwrites previous choices and updates the modified timestamp.
Exception flows	**E1 - Submission Window Closed (at Step 3)**: The server rejects a submission received at or after the configured cutoff or after draft generation has started. The system preserves the last saved declaration and displays a read-only view. **E2 - All Slots Unavailable (at Step 7)**: All slots are Unavailable. The system warns about potential shortages and saves the declaration if the Employee confirms. **E3 - Persistence Error (at Step 8)**: Database communication fails. System alerts the user and preserves current form entries for re-submission.
Business rules	1. Declarations apply strictly to un-drafted future weeks. 2. Undeclared weeks automatically inherit default baseline availability profiles. 3. "Unavailable" selections act as Hard Constraints; "Prefer Not" selections act as Soft Constraints penalizing solver objective scores.
Non-functional constraints	1. Strict privacy boundary: Employees can only view/edit their own declarations. 2. Server-side authorization follows NFR-04. 3. Matrix UI must be touch-friendly and fully operable on desktop and mobile viewports

7.4 Use Case UC-04

Use case ID	UC-04
Use case name	Configure Staffing Requirements
Brief description	The Shift Manager configures and manages weekly staffing requirement templates by defining required headcount per shift block (*Morning*, *Afternoon*, *Night*) and position/role. These staffing requirements define the target demand baseline for automated draft schedule generation (UC-05).
Primary actor	Shift Manager (Manager)
Secondary actors	None
Priority	Must
Related requirements	FR-04, FR-21 and FR-22. As an alternative to entering ordinary staffing demand, the Manager may post an overtime shift with its date, interval, work location, duties, required headcount, and optional informational compensation note. The offer is visible to Employees in the assigned team. Holiday status is stored separately from overtime classification.
Trigger	The Manager accesses the Staffing Configuration page to set up headcount demand for a target scheduling week or to update the default factory staffing template.
Pre-conditions	1. The Manager is authenticated with valid credentials and authorized for the production team (UC-01). 2. Active production positions/roles (e.g., Machine Operator, QC Inspector) are pre-loaded in the system configuration.
Post-conditions	**Success:** 1. Staffing requirements for the target week or baseline template are saved to the database. 2. Configured staffing demand is immediately accessible to the schedule generation engine (UC-05). 3. An audit log entry with timestamp and Manager ID is recorded. **Failure / Cancellation:** 1. No changes are saved to the database. 2. Existing baseline staffing requirements remain active.
Main flow	1. The Manager opens the Staffing Configuration screen (S-11). 2. The System retrieves and displays current weekly staffing demand templates and specific target week configurations. 3. The Manager selects a target week or chooses to edit the default weekly baseline template. 4. The System displays a matrix grid broken down by Day of Week (Mon–Sun), Shift Type (*Morning*: 06:00–14:00, *Afternoon*: 14:00–22:00, *Night*: 22:00–06:00), and Required Positions (*Machine Operator*, *QC Inspector*). 5. The Manager inputs or adjusts the required headcount values for each shift and position cell. 6. The Manager selects Save Staffing Configuration. 7. The System validates that all inputs are non-negative integers and comply with factory operational limits. 8. The System saves the updated staffing requirements to the database with a timestamp and audit record.

	9. The System displays a confirmation message indicating that staffing requirements are saved and ready for schedule generation (UC-05).
Alternative flows	**A1 - Load Default Baseline Template (at Step 3):** 1. The Manager selects "Load Default Template" for an unconfigured future week. 2. The System auto-populates the grid with standard baseline headcount values (e.g., 4 Operators per Morning shift, 2 per Night shift). 3. The Manager reviews or modifies the values and proceeds to Step 6. **A2 - Duplicate Requirements from Previous Week (at Step 3):** 1. The Manager selects "Copy from Previous Week". 2. The System fetches the exact staffing requirement matrix from the preceding week and populates the form. 3. The Manager reviews or modifies the values and proceeds to Step 6.
Exception flows	**E1 - Invalid Headcount Input (at Step 7):** The Manager enters negative numbers, non-numeric characters, or headcount exceeding total factory line capacity (e.g., > 50 per shift). The System highlights the invalid cells, displays an error message (*"Required headcount must be a Non-negative whole number within the configured limit"*), and blocks saving. **E2 - Modification Locked for Published Schedule (at Step 3):** The Manager attempts to edit the staffing configuration already attached to a published version (UC-05). The system opens the published version's configuration read-only. The Manager may save a separate configuration for a future schedule revision. **E3 - Database Persistence Error (at Step 8):** Database communication fails. The System alerts the Manager with an error message and preserves current form entries for re-submission.
Business rules	1. Only an authorized Manager may create or update staffing demand configurations for the assigned team. 2. Staffing requirements apply at the position/qualification level per shift block. 3. Each published version preserves the staffing configuration used at publication. New configurations may be used in a future revision through UC-05. 4. Zero is permitted where the staffing policy allows it. Any higher minimum for a critical position must be defined in the approved staffing configuration.
Non-functional constraints	1. Role Access Control follows NFR-04 for all protected write endpoints. 2. The performance targets in NFR-01 apply to the pages and operation specified there. 3. Usability: The matrix grid must be fully responsive and touch-friendly for both desktop and mobile viewports.

7.5 Use Case UC-05

Use case ID	UC-05
Use case name	Generate and Publish Weekly Schedule
Brief description	The Manager generates a weekly work schedule for the production team using staffing requirements, employee qualifications, availability, working-time constraints, and historical workload information. The Manager may manually adjust the draft schedule before publication. Planned changes to a published schedule are made through a new revision, preserving the previous version.
Primary actor	Manager
Secondary actors	None
Priority	Must
Related requirements	FR-04–FR-08, FR-22 and BR-07 (F02). The Manager may post an overtime shift and approve voluntary sign-ups before scheduling. Generation and manual editing consider only approved and eligible Employees. Publication revalidates current consent, approval and eligibility. The downstream schedule viewing and acknowledgement are specified by FR-09–FR-11 (F03).
Trigger	The Manager initiates the creation or planned revision of a weekly schedule.
Pre-conditions	- The Manager is authenticated and authorized to manage the production team. - An applicable staffing requirement template exists for the target week. - Employee qualifications, effective availability, and scheduling configuration are available. - Existing assignments needed to validate shift overlaps, rest periods, working-time limits, and applicable overtime conditions can be retrieved, including assignments across week boundaries. - The system can identify which historical weeks are available for fairness evaluation; complete four-week fairness history is not required for initial scheduling. - The target scheduling week is identified. When an Employee has no custom declaration, the applicable baseline availability is used. - Any assignment that requires overtime must have the information necessary to verify applicable overtime eligibility and consent before it can be treated as valid.
Post-conditions	**Success:** 1. A new official schedule version is published. 2. All assigned employees satisfy the applicable hard constraints. 3. Uncovered staffing positions, if any, remain clearly identified. 4. Only one published schedule version is current for the production team and week. 5. Previous published versions are preserved for traceability. 6. Affected employees can view the new assignments and changes through the schedule viewing functionality. **Failure or Cancellation:** 1. No invalid or partially published schedule becomes official. 2. The previous official version, if any, remains effective. 3. An existing draft may be retained for further editing.
Main flow	1. The Manager opens the Weekly Scheduling page. 2. The system displays the scheduling weeks and current schedule statuses for the Manager's assigned production team. 3. The Manager selects a target week for which no official schedule has yet been published. 4. The system retrieves a versioned input snapshot containing applicable staffing requirements, Employee qualifications, effective availability, existing assignments, working-time and rest rules, available workload history, and the current official schedule reference.

	5. The system validates the essential input data and identifies any missing information required for hard-constraint validation. 6. The Manager requests automatic schedule generation. 7. The scheduling engine generates a draft using the validated input snapshot. Every proposed assignment must satisfy hard constraints, including qualifications, effective Unavailable periods, non-overlapping assignments, applicable working-time limits, overtime eligibility, and required rest across week boundaries. The engine prioritizes staffing coverage before soft fairness and preference objectives. It does not claim globally optimal results. 8. The system displays the draft together with staffing coverage, uncovered worker-hours, scheduled-hour deviation from applicable Employee targets, night-shift imbalance, Prefer Not conflicts, and the number of historical weeks used. Missing historical weeks are explicitly identified and are not treated as zero-workload observations. 9. The Manager reviews the draft, warnings, and fairness indicators. The Manager may adjust the draft, save it, request regeneration, or select Publish Schedule. 10. Before publication, the system revalidates the draft against current input versions, existing assignments, applicable hard constraints, and the current official schedule version. Any stale or invalid draft is blocked from publication until reviewed or regenerated. A valid draft with staffing shortages requires explicit Manager confirmation. 11. When validation and any required shortage confirmation succeed, the system atomically publishes the new current official schedule version, preserves previous published versions, and updates the required acknowledgement records for affected Employees. 12. The system confirms the publication result and displays the new version information. 13. Authorized Employees, Managers, and Boss users can view the appropriate published schedule information through UC-06.
Alternative flows	**A1 – Manual Draft Adjustment (at Step 9)** 1. The Manager selects an assignment to modify. 2. The Manager changes an employee assignment or leaves an eligible position unfilled. 3. The system checks the proposed change against applicable hard constraints. 4. If valid, the system updates the draft and recalculates coverage and fairness indicators. 5. The Manager reviews the updated draft. 6. The flow returns to Step 9. **A2 – Staffing Shortage Warning (at Steps 9–10):** 1. The system identifies one or more uncovered staffing positions or intervals in an otherwise valid draft. 2. The system displays affected shifts, required staffing, assigned staffing, uncovered worker-hours, and the relevant shortage reasons when available. 3. The Manager reviews the warning. 4. If the Manager explicitly confirms publication with shortages, the system records the confirmation and continues with publication validation. 5. If the Manager declines, publication is cancelled. The draft remains available for further review, and the current official schedule remains unchanged. 6. The system must not relax a hard constraint to eliminate a staffing shortage. **A3 – Revise an Existing Published Schedule (at Step 3):**

	1. The system identifies the current official schedule version for the selected week. 2. The Manager selects Create Revision. 3. The system creates a revision draft based on the current published version. 4. The Manager changes only assignments for shifts that have not yet started, or requests regeneration of the eligible future portion. 5. The system validates the proposed changes and recalculates coverage and fairness indicators. 6. The Manager reviews the revised draft and requests publication. 7. The system performs the validations and any required shortage confirmation described in Steps 9–11. 8. The new version becomes the only current official version, while previous versions and previously effective assignments remain traceable. 9. Replacement during an ongoing shift is handled by UC-09 rather than by retroactive planned revision. **A4 – Save Draft Without Publishing (at Step 9)** 1. The Manager selects Save Draft. 2. The system saves the draft for later review. 3. No official schedule version is changed. 4. The use case ends. **A5 – Fairness or Preference Concern (at Step 9)** 1. The Manager reviews the fairness indicators and any assignments conflicting with Prefer Not preferences. 2. The Manager may accept the valid draft, adjust assignments, or request regeneration. 3. The system updates the draft when necessary. 4. The flow returns to Step 9.
Exception flows	**E1 – Missing or Invalid Input (at Step 5):** The system identifies missing staffing requirements or invalid essential scheduling data. It explains the problem and prevents generation until the data is corrected. **E2 – Invalid Manual Assignment (during A1 or A3):** A proposed assignment violates a hard constraint. The system rejects that change, explains the violation, and preserves the previous valid draft. **E3 – Solver Failure or Timeout (at Step 7):** The scheduling process fails or exceeds the configured execution limit without returning a validated draft. The system reports the failure and retains any previously valid saved draft. An unvalidated partial result must not be published. A valid draft containing staffing shortages is not considered a solver failure. **E4 – Outdated Draft Inputs (at Step 10):** Relevant staffing requirements, availability, qualifications, existing assignments, or other hard-constraint inputs have changed since draft generation. The system prevents publication and requires the Manager to review or regenerate the draft against current information. **E5 – Conflicting Official Version (at Step 11):** Another official schedule version has been published since the draft was created or last validated. The system prevents the outdated draft from overwriting the current official version and requires another review. **E6 – Publication Transaction Failure (at Step 11):** The schedule version and required acknowledgement updates cannot be committed consistently. The system reports the failure, preserves the previously current official schedule and its associated acknowledgement state, and does not expose a partially published result.

Business rules	1. Only an authorized Manager may generate, adjust, and publish schedules for the assigned production team. 2. Every assignment must satisfy applicable hard constraints, including qualifications, effective Unavailable periods, non-overlapping shifts, working-time and overtime eligibility, and required rest across week boundaries. 3. Scheduling prioritizes staffing coverage over soft fairness and preference objectives. Hard constraints must never be relaxed to improve coverage. 4. The fairness evaluation uses the target scheduling week and up to four preceding official weeks. Missing history must be identified and must not be counted as zero workload. 5. Workload evaluation includes scheduled-hour deviation from applicable Employee targets, night-shift imbalance among comparable eligible Employees, and Prefer Not conflicts. 6. A valid schedule with uncovered staffing positions may be published only after explicit Manager confirmation. 7. A Manager may manually adjust drafts, but every change must be revalidated against hard constraints. 8. Publication must validate current input versions and prevent stale drafts from silently overwriting newer official schedules. 9. Only one official schedule version may be current for a given team and week. Previously published versions must be retained. 10. Planned revisions may change only future assignments. Operational replacement during an ongoing shift is handled by UC-09. 11. Publication must consistently update the official schedule version and the required acknowledgement state of affected Employees. 12. Overtime assignments require validation under the applicable legal rules and approved business policy; the five-week fairness window does not substitute for required overtime history.
Non-functional constraints	- Role-based access must prevent unauthorized schedule creation and publication. - Publication must not leave incomplete or contradictory official schedule data. - Published changes must be traceable to the responsible Manager. - Constraint violations and staffing shortages must be clearly understandable to the Manager. - Measurable performance and reliability targets must be aligned with Section 5.2.

7.6 Use Case UC-06

Use case ID	UC-06
Use case name	View Published Schedule and Changes
Brief description	An authenticated user views the current official (published) schedule within their authorization. An Employee sees their own assigned shifts for a selected week, with new, changed, and cancelled assignments highlighted relative to the previous published version. A Manager sees the team schedule together with acknowledgement progress, and a Boss sees read-only staffing and workload summaries.
Primary actor	Employee
Secondary actors	None. The Manager and the Boss act as the primary actor in alternative flows A2 and A3.

Priority	Must
Related requirements	F03 – Centralized Schedule & Change Acknowledgement. See the FR mappings in Sections 6.3 and 9.
Trigger	The user opens the schedule page of their role (My Schedule, Team Schedule, or Staffing Overview) or selects a different week.
Pre-conditions	The user is authenticated (UC-01) and belongs to the production team. - Schedules are published through UC-05 or UC-09. If no schedule exists for the selected week, the system handles this in E1.
Post-conditions	**Success:** 1. The user has seen the current official schedule version for the selected week, limited to what their role may view. 2. For an Employee, the changes since their last acknowledged applicable version are available. 3. No schedule data, assignment, or acknowledgement status has been modified. **Failure or Cancellation:** 1. No draft, outdated, or unauthorized data is shown as the official schedule. 2. Schedule data remains unchanged.
Main flow	1. The Employee opens the My Schedule page. 2. The system identifies the Employee and the current week. 3. The system retrieves the current official schedule version for the Employee's production team and the Employee's assignments for that week. 4. The system displays the Employee's assigned shifts with date, shift, start and end time, and position. 5. The system compares consecutive published versions and labels New, Changed, and Cancelled assignments in text. On first publication it labels the Employee's assignments New. It also shows cumulative changes since the last acknowledged applicable version and provides access to intermediate versions. 6. The system displays the version number, the publication time, the total scheduled hours and night shifts for the week, and the acknowledgement status of the latest applicable update. 7. If an update is awaiting acknowledgement, the system displays a notice with Review & Acknowledge, which leads to UC-07. 8. The Employee reviews the schedule and may select a change indicator to see its details. 9. The system shows the previous and the current assignment for the selected change.
Alternative flows	**A1 – Select Another Week (at Step 4)** 1. The Employee selects the previous or the next week. 2. The system returns to Step 3 for the selected week. **A2 – Manager Views the Team Schedule (at Step 1)** 1. The Manager opens the Team Schedule page. 2. The system displays the current official version for the Manager's production team: each employee's assignments by day, total hours, and uncovered staffing positions. 3. The system displays, for the latest published update, which employees have acknowledged it and which are still pending. 4. The Manager may filter the list to pending acknowledgements only or select another week. **A3 – Boss Views the Staffing and Workload Overview (at Step 1)** 1. The Boss opens the Staffing Overview page.

	2. The system displays read-only summaries for the selected week: staffing coverage, uncovered positions, absences, pending acknowledgements, and workload distribution (scheduled hours and night shifts per employee). 3. The Boss may select another week. The Boss cannot modify schedules. **A4 – No Assigned Shifts (at Step 4)** 1. The Employee has no assignments in the selected week and no pending applicable update. 2. The system displays a message that no shifts are assigned for that week, together with the version information. 3. The use case ends. **A5 – No Changes Since the Previous Version (at Step 5)** 1. A previous version exists and none of the Employee's assignments differ from it. 2. If there are no changes since the previous version, the system continues at Step 6. It still makes any outstanding applicable update available for review.
Exception flows	**E1 – No Published Schedule (at Step 3):** No official schedule exists for the selected week. The system tells the user that no schedule has been published yet and does not show draft data to an Employee. For a Manager, the page offers a link to Weekly Scheduling (UC-05). **E2 – Unauthorized Access (at Step 3):** The user requests another employee's roster or a page that the role does not permit. The system denies the request, shows a "no permission" message, and discloses no data. **E3 – Data Retrieval Failure (at Step 3):** The schedule cannot be loaded. The system shows an error message with a retry option and does not present partial or stale data as the official schedule. **E4 – Session Expired (at any step):** The session is no longer valid. The system redirects the user to the login page (UC-01) and, after login, returns the user to the requested page. **E5 – Schedule Updated While Viewing (at Steps 4–9):** A newer version is published after the page was loaded. The system indicates that a newer version is available and, when the user refreshes, continues again from Step 3.
Business rules	1. Only the current official (published) version is presented as the schedule. Drafts are visible only to the Manager (UC-05). 2. An Employee may view only their own assignments and the changes that affect them; a Manager may view the schedules of their own production team; a Boss has read-only access to authorized staffing and workload summaries. 3. A change is a new, modified, or cancelled assignment between two consecutive published versions that affects the Employee, including changes caused by replacement coverage (UC-09). 4. Change indicators identify the type of change by a text label as well as by colour. 5. Viewing a schedule does not count as an acknowledgement; acknowledgement requires the explicit action in UC-07. 6. Uncovered staffing positions remain visible to the Manager and the Boss until they are resolved. 7. Previous published versions are retained, but only the current version is presented as official.
Non-functional constraints	Role-based access control must be enforced on the server for every schedule request. The page must show one consistent schedule version, never a mixture of versions. The schedule must be readable on desktop and mobile browsers. Change indicators and statuses must not rely on colour alone. - Page load time targets follow Section 5.2.

7.7 Use Case UC-07

Use case ID	UC-07
Use case name	Acknowledge Schedule Update
Brief description	An Employee confirms that they have seen a newly published schedule or a schedule change that affects them, including a change that results from replacement coverage. The system records the acknowledgement against the specific schedule version, so that the Manager can see which employees have acknowledged the update and which are still pending.
Primary actor	Employee
Secondary actors	Manager (views the acknowledgement status)
Priority	Must
Related requirements	F03 – Centralized Schedule & Change Acknowledgement; F04 – Absence Reporting & Shift Coverage (updates produced by UC-09). See the FR mappings in Sections 6.3 and 9.
Trigger	The Employee sees a notice of a schedule update awaiting acknowledgement and selects Review & Acknowledge.
Pre-conditions	- The Employee is authenticated (UC-01). - A new schedule version has been published (UC-05 or UC-09) that adds, modifies, or cancels at least one of the Employee's assignments. - The Employee's acknowledgement of that version is Pending. - The version is the latest applicable update for the Employee in the selected week. A newer version that does not affect the Employee does not supersede it.
Post-conditions	**Success:** 1. The acknowledgement is recorded with the Employee, the schedule version, and the time of acknowledgement. 2. The update status for the Employee changes from Pending to Acknowledged, and the pending notice is removed. 3. The Manager can see the Employee's acknowledgement status in the Team Schedule view. 4. The schedule and the Employee's assignments are unchanged. **Failure or Cancellation:** 1. No acknowledgement is recorded and the update remains Pending. 2. The schedule and the Employee's assignments are unchanged.
Main flow	1. The Employee opens the My Schedule page (UC-06). 2. The system displays a notice that one or more schedule updates are awaiting acknowledgement. 3. The Employee selects Review & Acknowledge for an update. 4. The system displays the update summary: week, schedule version, publication time, and the Employee's cumulative new, changed, and cancelled assignments since the last acknowledged applicable version, with access to intermediate history. 5. The Employee reviews the changes, confirms that they have been reviewed, and selects Acknowledge. 6. The system checks that the update is still the latest version affecting the Employee and that its status is still Pending.

	7. The system records the acknowledgement with the Employee, the schedule version, and the current time. 8. The system changes the status to Acknowledged, removes the pending notice, and confirms the result to the Employee. 9. The system makes the acknowledgement status visible to the Manager in the Team Schedule view (UC-06, A2).
Alternative flows	**A1 – Updates Are Pending for Several Weeks (at Step 3)** 1. The system lists the latest pending applicable update for each affected week, with its version and publication time. 2. The Employee selects one update and continues at Step 4. 3. After acknowledging it, the Employee returns to the list and handles the remaining updates in the same way. **A2 – Update Superseded by a Newer Version (at Step 6)** 1. The system detects that a newer version affecting the Employee was published after the update was displayed. 2. The system marks the earlier update as Superseded; it is not recorded as Acknowledged. 3. The system displays the newer update and returns to Step 4. **A3 – Manager Follows Up on Pending Acknowledgements (after Step 9)** 1. The Manager opens the Team Schedule page (UC-06, A2). 2. The system shows, for the latest version, which employees have acknowledged it and which are still pending. 3. The Manager may filter to pending acknowledgements and follow up with those employees outside the application. **A4 – Employee Leaves Without Acknowledging (at Step 5)** 1. The Employee selects Back to My Schedule. 2. The system keeps the update Pending and keeps the notice visible. 3. The use case ends.
Exception flows	**E1 – Already Acknowledged (at Step 6):** The action is repeated (for example, by a double click or from a second device) after the acknowledgement was already recorded. The system does not create a duplicate record, shows the existing acknowledgement time, and continues at Step 8. **E2 – Update Not Available to This Employee (at Step 4):** The requested update does not exist or does not affect this Employee. The system denies access, shows a message, and discloses no data. **E3 – Recording Failure (at Step 7):** The acknowledgement cannot be saved. The system reports the failure, keeps the update Pending, and lets the Employee try again. **E4 – Session Expired (at Step 5 or Step 7):** The session is no longer valid. The system redirects the Employee to the login page (UC-01) without recording an acknowledgement. After login, the pending update is displayed again.
Business rules	1. Only the affected Employee may acknowledge their own update; neither the Manager nor the Boss can acknowledge on an Employee's behalf. 2. An acknowledgement applies to one specific schedule version; acknowledging an earlier version does not cover a later one.

	3. Acknowledgement only confirms that the Employee has seen the update. It does not accept, reject, or change assignments; absences are reported separately (UC-08), and shifts change only through the managed coverage workflow (UC-09). 4. A published update takes effect without waiting for acknowledgements. 5. At most one acknowledgement is recorded per Employee and schedule version, and users cannot edit or delete it. 6. The Manager may view the acknowledgement status of the production team; an Employee sees only their own status.
Non-functional constraints	- The acknowledgement must be recorded exactly once, even if the action is repeated or performed concurrently. - The acknowledgement record must be traceable to the Employee, the schedule version, and the time. - Use text to distinguish Pending, Acknowledged, Superseded, and Not Required where applicable. - The Acknowledge action must be operable by keyboard and usable on mobile browsers. - Acknowledgement reliability follows NFR-05. My Schedule page load time follows NFR-01.

7.8 Use Case UC-08

Use case ID	UC-08
Use case name	Report Absence
Brief description	An Employee reports an absence for their own future or ongoing assigned shift. The system records the report and creates or reuses an unresolved coverage request for Manager action.
Primary actor	Employee
Secondary actors	Manager (reviews the coverage request in UC-09)
Priority	Must
Related requirements	FR-13, FR-02, BR-08
Trigger	An Employee determines they cannot work all or part of their assigned shift before it ends and selects Report Absence.
Pre-conditions	1. Employee is authenticated (UC-01). 2. Target shift is assigned to the Employee in an active published schedule version. 3. Target shift has not ended. 4. If an unresolved coverage request already exists, the system reuses it.
Post-conditions	**Success**: 1. Absence report is saved with reason code, optional details, and timestamp. 2. Shift assignment status transitions to "Absence Reported". 3. An unresolved coverage request is created or reused for Manager resolution (UC-09). 4. The original assignment remains traceable. The affected interval is shown as uncovered until valid coverage is confirmed. **Failure / Cancellation**: 1. No absence report is generated. 2. Shift assignment remains in its original active state.
Main flow	1. Employee opens the Report Absence screen (S-06).

	2. The system lists the Employee's future and ongoing assigned shifts from the current published roster. 3. Employee selects the affected shift. 4. Employee selects a reason (Illness, Family Emergency, Other) and enters optional notes. 5. Employee clicks **Submit report**. 6. The server validates shift ownership, timing, and the required reason category, and checks for an existing coverage request. 7. The system records the absence and creates or reuses the unresolved coverage request. 8. System updates the Manager's Absence & Coverage dashboard (S-07). 9. The system confirms the report and shows coverage status. The original assignment remains traceable and no substitute is assigned by the report.
Alternative flows	**A1 - Review Past Absence Reports (at Step 2)**: 1. Employee chooses to view historical absence reports. 2. System renders report histories and status outcomes (Resolved - covered / Unresolved). **A2 - Emergency Mid-Shift Absence (at Step 3)**: 1. Employee reports an absence for an ongoing shift that has started but not ended. 2. System logs the actual report timestamp and flags an urgent mid-shift coverage request for the Manager (UC-09).
Exception flows	**E1 - Shift Already Ended (at Step 6)**: Target shift ended in the past. System rejects the submission with error: "Cannot report absence for completed shifts". **E2 - Duplicate Absence Report (at Step 6)**: The system displays the existing report and coverage status without creating a duplicate coverage request. **E3 - Schedule Version Updated (at Step 6)**: Manager published a new schedule revision modifying the shift. System notifies the user of updated roster data and refreshes shift choices. **E4 - Transaction Failure (at Step 7)**: Database write fails. System alerts user and retains input entries for re-submission.
Business rules	1. Submitting an absence report does not alter shift ownership; assignment transfers only upon Manager replacement confirmation in UC-09. 2. Employees can only report absences for their own assigned shifts. 3. The system retains the reporting actor and timestamp for audit and shows only authorized absence summaries on the Boss dashboard (S-08).
Non-functional constraints	1. Server must validate JWT claims to prevent unauthorized shift manipulation. 2. Authorization and accessible interaction follow NFR-04 and NFR-03. 3. UI must clearly distinguish absence reporting from arbitrary shift cancellation.

7.9 Use Case UC-09

Use case ID	UC-09
Use case name	Arrange Replacement Coverage
Brief description	The Manager arranges replacement coverage for an assigned employee who reports an absence or is recorded as a no-show. The system identifies eligible candidates, allows the Manager to send a replacement offer, and records the candidate's response. A replacement becomes official only after employee acceptance and Manager confirmation. The updated assignment is recorded through the published schedule revision mechanism.
Primary actor	Manager
Secondary actors	Employee (replacement candidate)
Priority	Must
Related requirements	FR-14, FR-15, FR-16, FR-17, FR-18 (F04); related published-schedule visibility and acknowledgement are specified by FR-09–FR-11 (F03).
Trigger	An assigned employee reports an absence, or the Manager identifies a no-show requiring replacement coverage.
Pre-conditions	- The Manager is authenticated and authorized to manage the production team. - An official schedule exists and the affected assignment can be identified. - The shift requiring coverage has not ended. - Employee qualification, availability, and current assignment data are accessible. - The affected staffing position has not already been resolved by another confirmed replacement.
Post-conditions	**Success:** 1. An eligible replacement candidate has accepted an offer and the Manager has confirmed it. 2. The replacement is effective only for the confirmed valid interval. 3. The offer is Confirmed, and the associated coverage request records the resulting coverage outcome. 4. The current official schedule version and required acknowledgement records are consistent. 5. Previous versions, previously effective intervals, candidate responses, and Manager decisions remain traceable. 6. Any historical shortage before the replacement effective time remains recorded. **Failure or Cancellation:** 1. No unconfirmed replacement is made official. 2. The coverage request remains Unresolved unless it is legitimately cancelled or the shift has ended. 3. Cancelled or Expired offers cannot create official assignments. 4. Transaction failure does not leave a partially updated official schedule. 5. A request remaining uncovered at shift end is retained as Unresolved at Shift End, not Resolved.
Main flow	1. The Manager opens the Absence and Shift Coverage page. 2. The system displays absence reports, recorded no-shows, and unresolved coverage requests. 3. The Manager selects an unresolved request associated with a valid published assignment.

	4. The system identifies the affected staffing position, the effective interval requiring coverage, the current official schedule version, and any existing active offer. 5. The system retrieves eligible replacement candidates. Eligibility checks include qualifications, effective availability, existing assignments, ordinary working-time and overtime conditions, applicable consent, required rest before and after the proposed interval, and the current schedule version. The absent Employee is excluded as a replacement for their own unresolved absence. 6. If the position has no active Pending or Accepted offer, the Manager selects an eligible candidate and sends a replacement offer. 7. The system records the offer as Pending, links it to the uncovered position, and makes it available only to the addressed candidate. 8. The addressed Employee reviews the offer and selects Accept. An overtime replacement offer shows the remaining interval, duties, location and overtime terms and requires explicit consent. At confirmation the system verifies that consent covers the final interval and terms (FR-22 and BR-07). 9. The system validates the response and changes the offer state from Pending to Accepted. 10. The Manager reviews the accepted offer and selects Confirm Replacement. 11. The system revalidates the candidate's eligibility, consent where applicable, working-time and rest limits, affected staffing position, current official schedule version, effective replacement interval, and unresolved coverage state. 12. The system atomically records the effective assignment change, changes the offer to Confirmed, updates the coverage request, publishes the corresponding current official schedule version, preserves previous version history, and creates the required acknowledgement updates. 13. The system confirms the outcome. Any historical uncovered interval remains traceable, and authorized users can view the updated coverage and schedule information.
Alternative flows	**A1 – Manager Records a No-show (at Step 3):** 1. After the scheduled shift has started, the Manager identifies an assigned Employee who has not reported for duty. 2. The Manager records a no-show with the observation timestamp. 3. The system verifies the assignment and checks whether an unresolved coverage request already exists for the same affected assignment. 4. If an unresolved request exists, the system reuses it rather than creating a duplicate. 5. Otherwise, the system creates an unresolved coverage request. 6. The system records the responsible Manager and makes the affected coverage requirement visible. 7. The flow continues at Step 4. **A2 – Candidate Rejects the Offer (at Step 8)** 1. The employee rejects the replacement offer. 2. The system records the rejection and informs the Manager through the application. 3. The Manager may select another eligible candidate. 4. The flow returns to Step 6, or ends with the position unresolved. **A3 – No Eligible Candidate Is Available (at Step 5)** 1. The system reports that no currently eligible replacement candidate was found.

	2. The affected position remains uncovered. 3. The Manager may review the relevant data or end the process without confirmation. 4. The system retains the unresolved coverage status. **A4 – Candidate Does Not Respond (at Step 8)** 1. The offer remains Pending. 2. The Manager may wait or cancel the pending offer. 3. If cancelled, the system records the cancellation. 4. The Manager may send a new offer to another candidate. 5. An unanswered offer never creates an official assignment. **A5 – Manager Does Not Confirm (at Step 10)** 1. The Manager declines to confirm the accepted offer and cancels it. 2. The system records the decision and makes the cancellation visible to the candidate. 3. No replacement assignment is published. 4. The staffing position remains unresolved. **A6 – Replacement During an Ongoing Shift (at Step 11):** 1. The system determines that the affected shift has started but has not ended. 2. The system validates the proposed replacement for the remaining shift interval only. 3. If the Manager confirms the replacement at time T, the replacement becomes effective no earlier than T. 4. Previously effective assignment intervals and any uncovered time before T remain in historical records. 5. The flow continues with the atomic confirmation in Step 12.
Exception flows	**E1 – Invalid Absence or Assignment (at Step 4):** The absence/no-show cannot be linked to a valid scheduled assignment. The system reports the problem and prevents replacement confirmation. **E2 – Candidate Becomes Ineligible (at Step 11):** The candidate no longer satisfies an applicable eligibility or hard constraint. The system rejects confirmation, records the reason, cancels the invalid offer when required, and keeps the coverage request Unresolved so that another candidate may be considered. **E3 – Coverage Already Resolved (at Step 11):** The staffing position has already been covered by another confirmed action. The system prevents duplicate confirmation. **E4 – Conflicting Schedule Revision (at Step 11):** The official schedule changed after the offer was created. The system requires revalidation against the latest official version before confirmation. **E5 – Shift Has Ended (at Step 11):** The affected shift has ended before confirmation. The system rejects retroactive replacement, closes any active offer as Expired where applicable, and records the unresolved coverage outcome as Unresolved at Shift End. Previously effective assignment history is preserved. **E6 – Transaction Failure or Competing Confirmation (at Step 12):** Another confirmation or a processing failure prevents a consistent update. The system does not create a duplicate assignment. Failed transactions roll back all related changes, and the previously committed official schedule and coverage records remain consistent.
Business rules	1. Only an authorized Manager may issue and confirm replacement coverage for the assigned production team. 2. Only the addressed Employee may accept or decline a Pending offer.

	3. Offer states are Pending, Accepted, Declined, Cancelled, Expired, and Confirmed. 4. A Pending offer may become Accepted or Declined through the candidate's response. A Pending or Accepted offer may be Cancelled by an authorized Manager or Expired when the applicable shift ends. Only an Accepted offer may become Confirmed after eligibility revalidation and Manager approval. 5. Declined, Cancelled, Expired, and Confirmed offers are terminal. Terminal offers cannot be accepted or confirmed again. 6. Only one Pending or Accepted offer may exist for the same uncovered staffing position. Cancellation or expiry releases this restriction. 7. Coverage request states are separate from offer states: Unresolved, Resolved, Cancelled, and Unresolved at Shift End. Resolved must not be used as an offer state. 8. A coverage request may be Cancelled only when the actual staffing requirement is legitimately removed through a valid schedule revision, not merely because a replacement was not found. 9. A Manager may record a no-show only after the shift has started. Duplicate absence or no-show events must not create duplicate unresolved coverage requests. 10. The absent Employee is not an eligible replacement candidate for their own unresolved absence. 11. Replacement confirmation must revalidate qualifications, effective availability, working-time and overtime conditions, applicable consent, rest requirements before and after the replacement, the current official schedule version, and the unresolved staffing position. 12. Confirmation must consistently update the assignment, offer state, coverage outcome, published schedule version, and required acknowledgement records. 13. Replacement during an ongoing shift takes effect only for the confirmed remaining interval. Earlier effective intervals and shortages must remain traceable. 14. A request that remains uncovered at shift end must be recorded as Unresolved at Shift End and must not be counted as successfully covered.
Non-functional constraints	1. Role and team authorization must be enforced by the backend for offer issuance, response, and confirmation. 2. Offer responses and Manager decisions must record the responsible actor, timestamp, and relevant reason where applicable. 3. Repeated or concurrent confirmation attempts must not create duplicate assignments, offers, or inconsistent official schedule versions. 4. The application must display offer states and coverage request states separately using textual indicators, not colour alone. 5. Acceptance and rejection actions must be accessible by keyboard and usable on mobile browsers. 6. Replacement confirmation and publication must preserve transactional consistency, and failed operations must not expose partial committed results. 7. Applicable measurable performance requirements shall follow Section 5.2.

8 Prototype / Mockups

Screen	*Related use case(s)*	*Mockup / link*
S-01 – Login	UC-01	Figure 8.1
S-02 – My Schedule (Employee)	UC-06, UC-07	Figure 8.2
S-03 – Schedule Update and Acknowledgement (Employee)	UC-07	Figure 8.3
S-04 – Weekly Scheduling (Manager)	UC-05	Figure 8.4
S-05 – Team Schedule and Acknowledgement Status (Manager)	UC-06 (A2), UC-07 (A3)	Figure 8.5
S-06 – Report Absence and Replacement Offers (Employee)	UC-08, UC-09	Figure 8.6
S-07 – Absence and Shift Coverage (Manager)	UC-09	Figure 8.7
S-08 – Staffing and Workload Overview (Boss)	UC-06 (A3)	Figure 8.8
S-09 – Employee Account Management (Manager)	UC-02	Figure 8.9
S-10 – Availability and Overtime Sign-up (Employee)	UC-03	Figure 8.9
S-11 – Staffing Configuration and Overtime Posting (Manager)	UC-04, UC-05	Figure 8.9

Figure 8.1 – S-01 Login (UC-01)

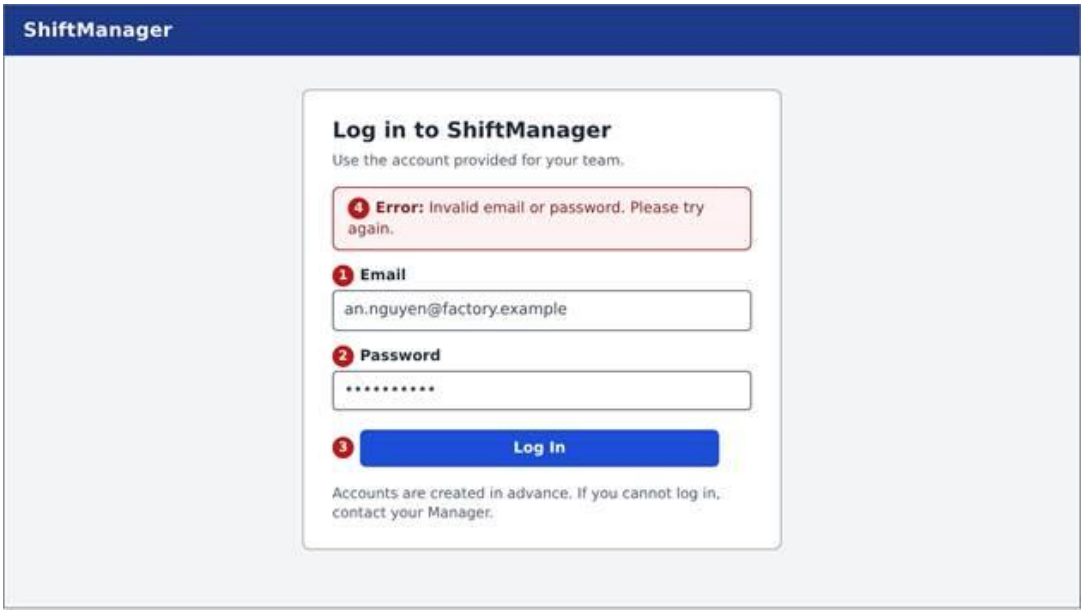


Figure 8.2 – S-02 My Schedule (Employee) (UC-06, UC-07)

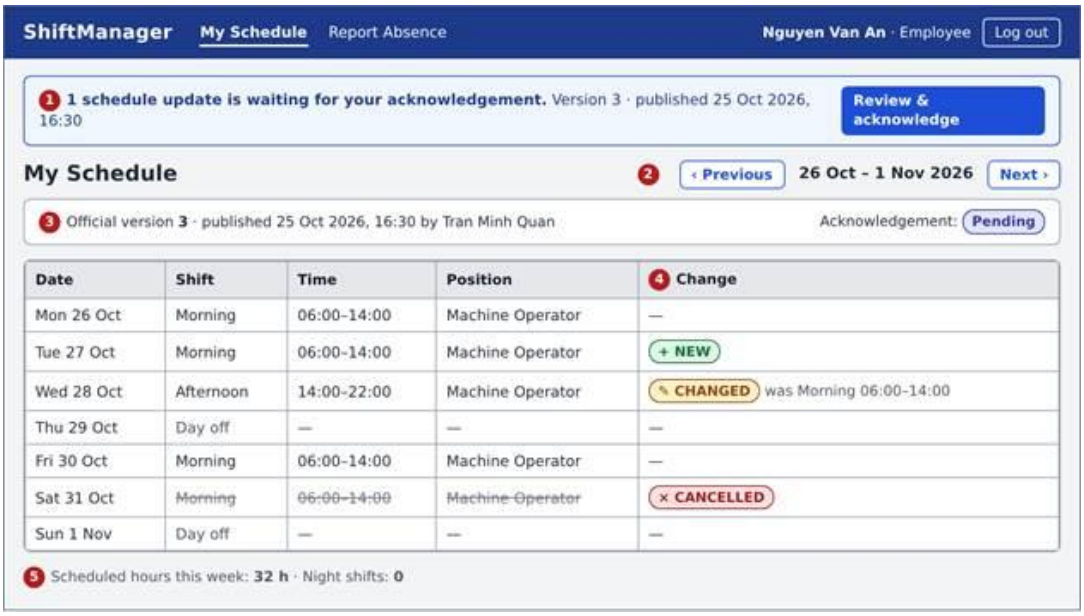


Figure 8.3 – S-03 Schedule Update and Acknowledgement (Employee) (UC-07)

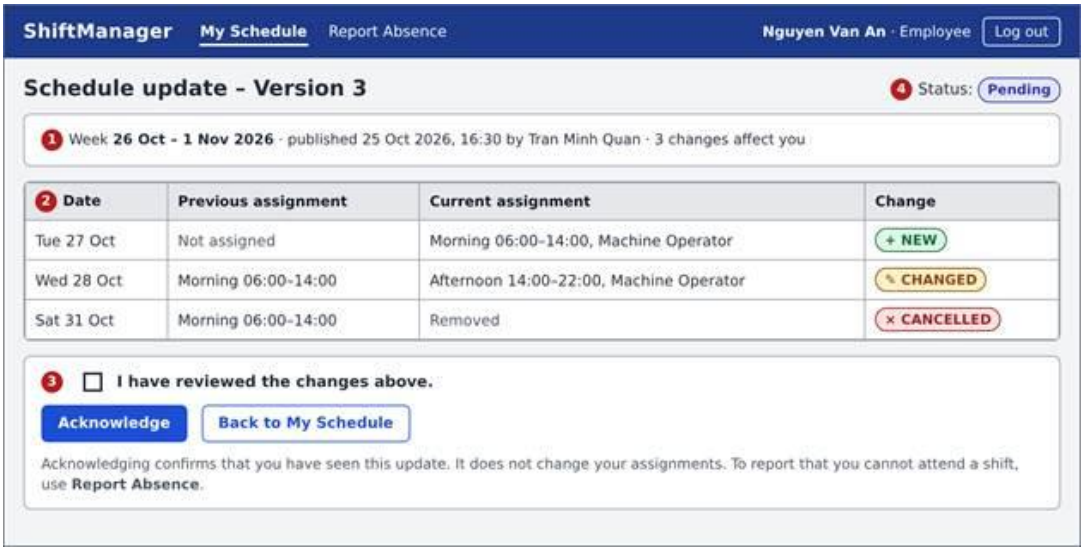


Figure 8.4 – S-04 Weekly Scheduling (Manager) (UC-05)

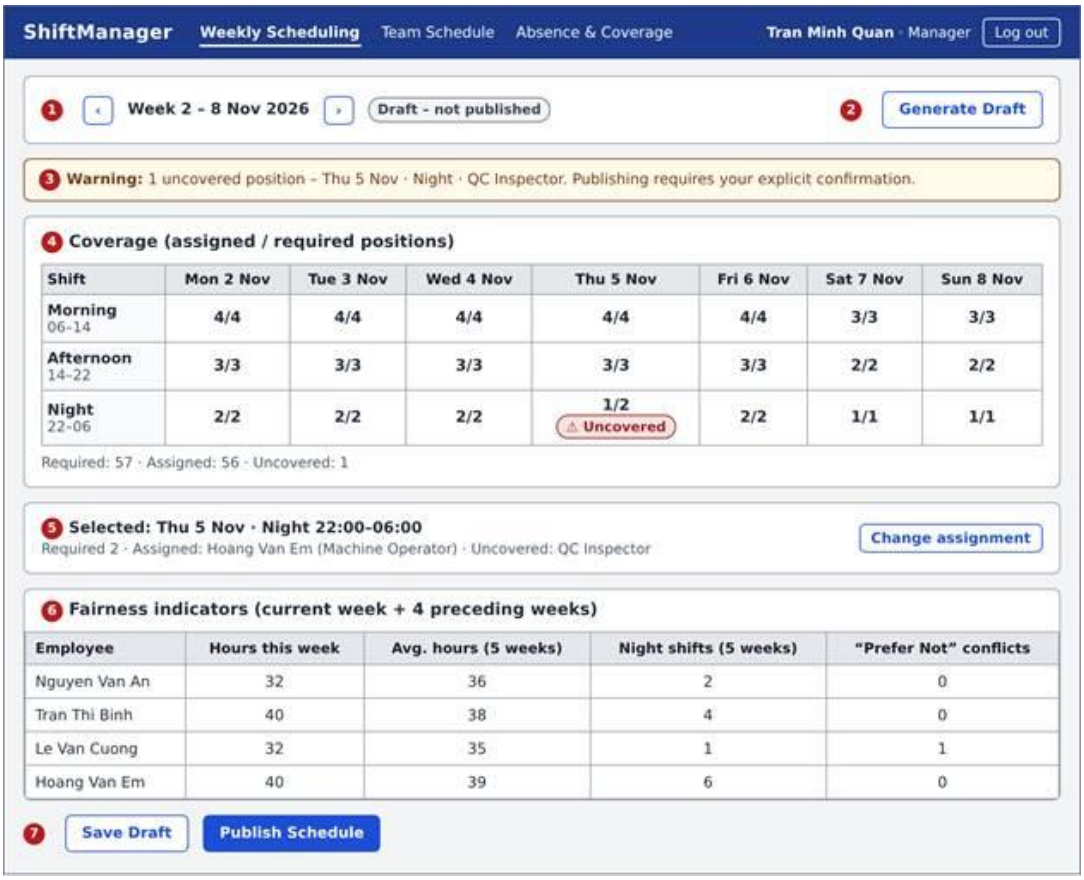


Figure 8.5 – S-05 Team Schedule and Acknowledgement Status (Manager) (UC-06 (A2), UC-07 (A3))

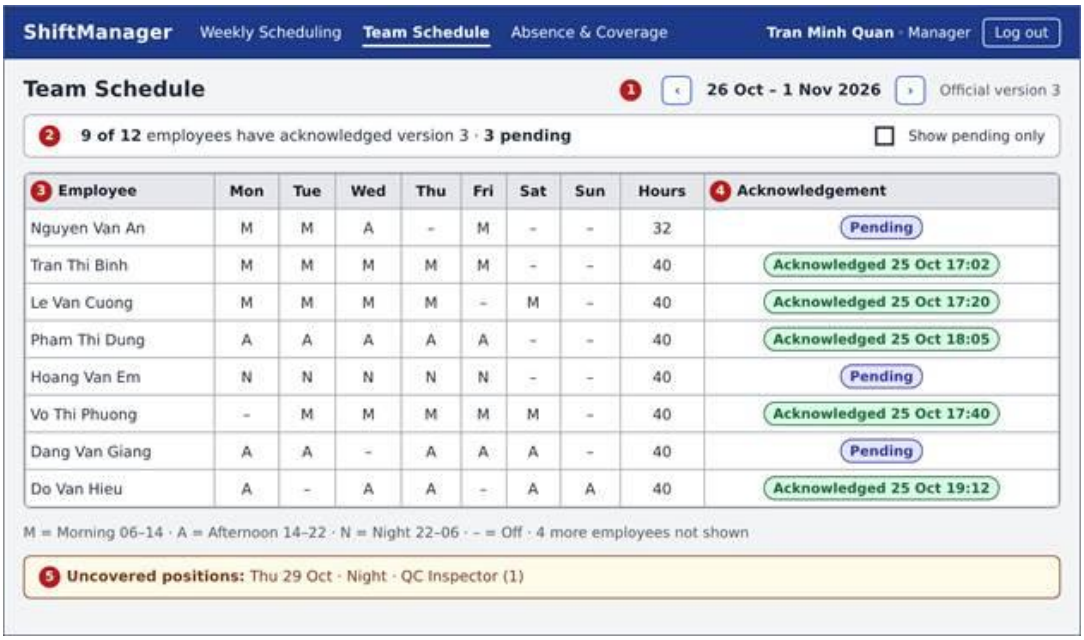


Figure 8.6 – S-06 Report Absence and Replacement Offers (Employee) (UC-08, UC-09)

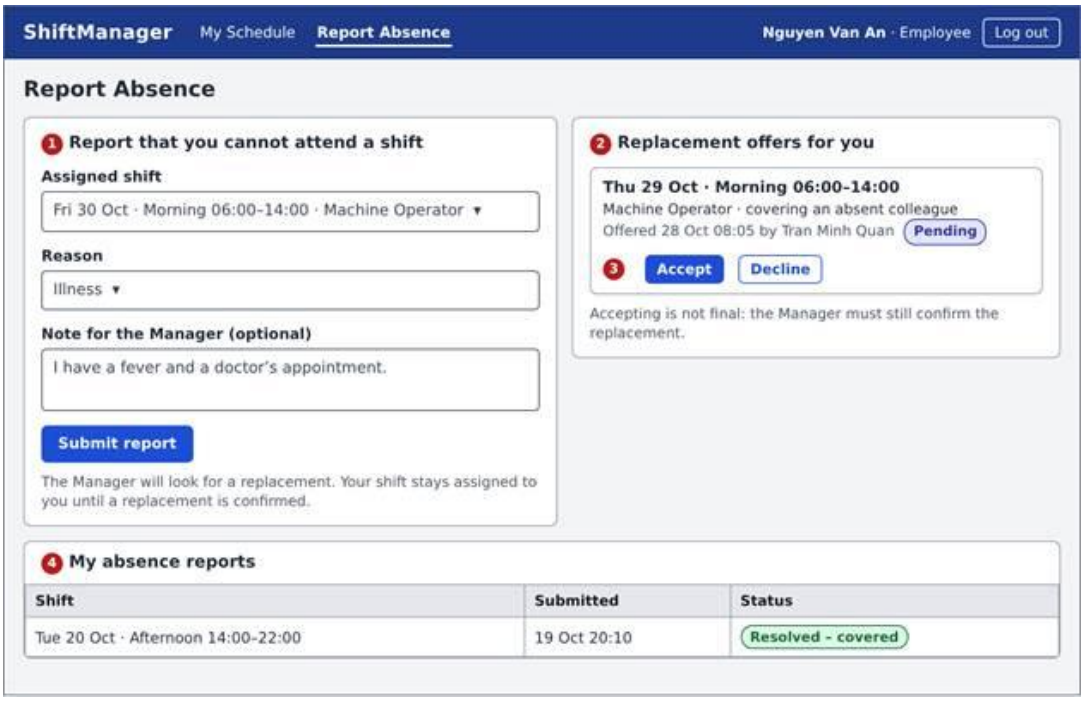


Figure 8.7 – S-07 Absence and Shift Coverage (Manager) (UC-09)

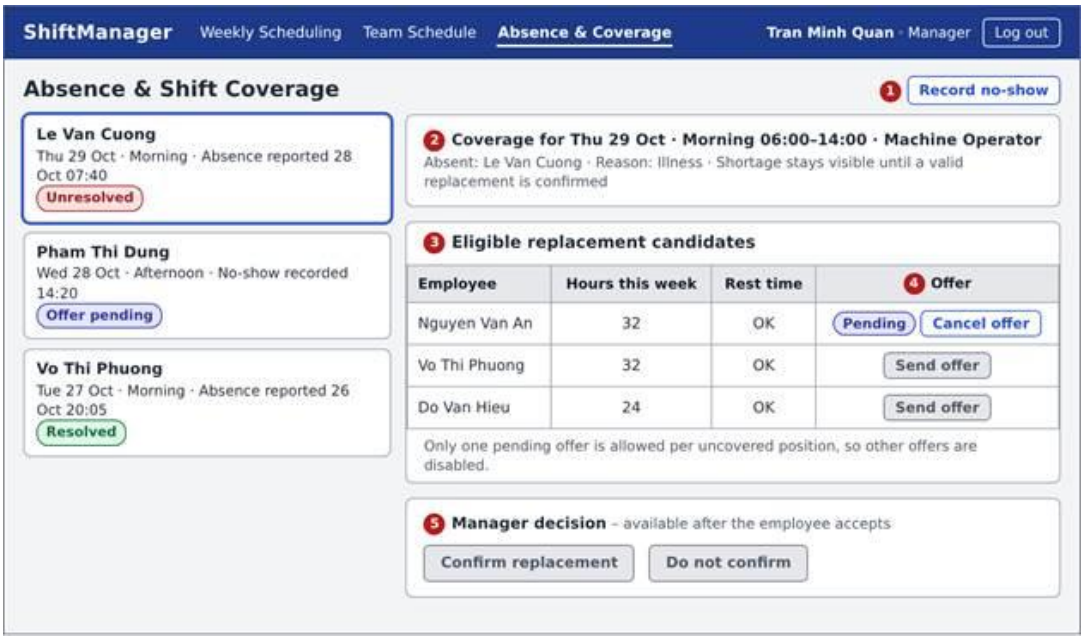


Figure 8.8 – S-08 Staffing and Workload Overview (Boss) (UC-06 (A3))



[Open the ShiftManager interactive prototype (S-01–S-11)]

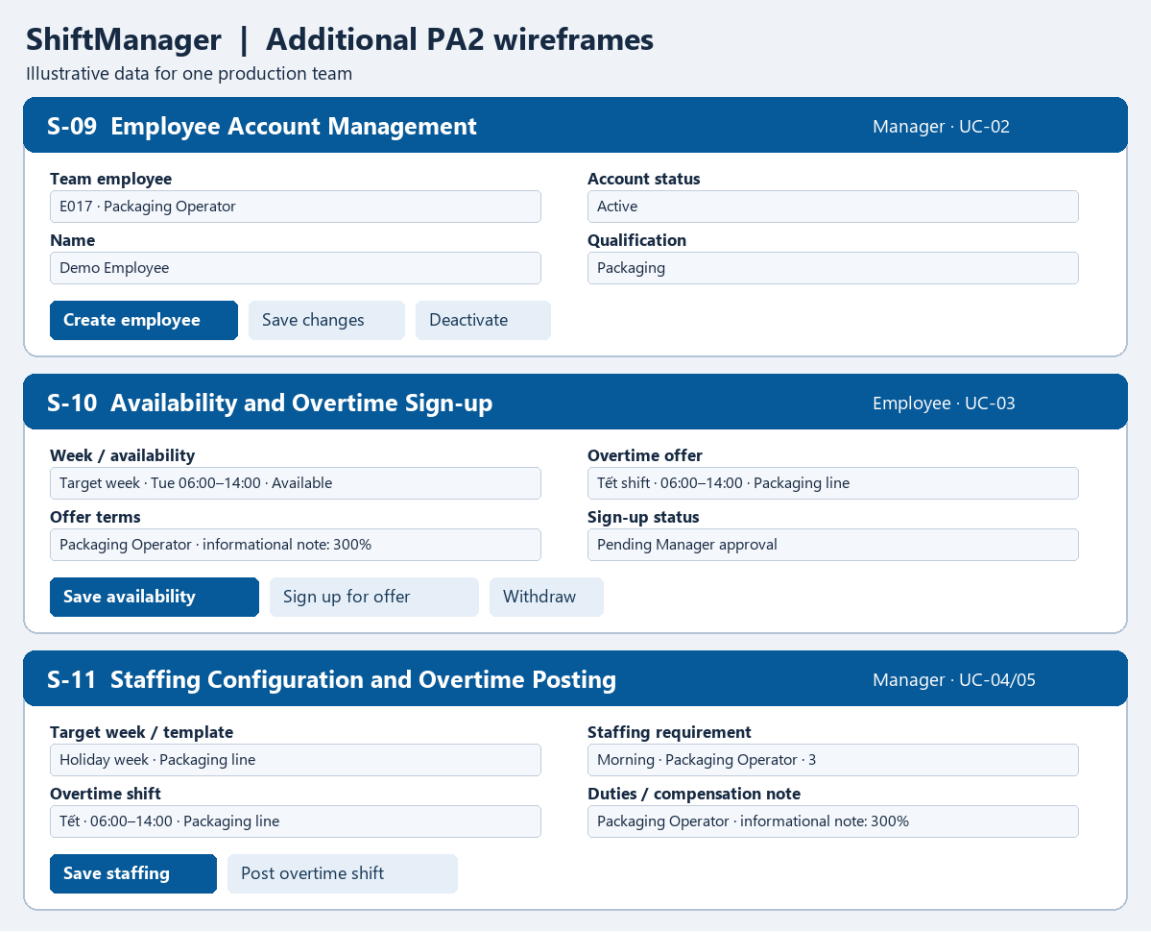
S-09 Employee Account Management (UC-02, FR-19). The Manager sees team accounts and status. Create opens the employee form. Edit changes permitted fields. Activate and Deactivate require confirmation and show the outcome.

S-10 Availability and Overtime Sign-up (UC-03, FR-20/22). The Employee selects a week and edits availability. A separate list displays overtime terms, Sign up or Withdraw, and approval status. Availability and consent are recorded separately.

S-11 Staffing Configuration and Overtime Posting (UC-04/05, FR-21/22). The Manager sets headcounts and can post an overtime shift with its interval, location, duties and informational compensation note. S-04 displays sign-ups and eligibility before approval and publication.

S-06 and S-07 overtime replacement. Show the remaining interval and terms to the candidate. Require explicit overtime consent where applicable. After confirmation, display the effective interval.

Figure 8.9 – Additional wireframes for S-09, S-10 and S-11



9 Requirements Traceability Matrix

Req. ID	Use case	Screen	Design element (PA3)	Test cases (PA4)
FR-01	UC-01	S-01	Pending (PA3)	Pending (PA4)
FR-02	UC-01; UC-02; authorization applies to protected UC operations	S-01; S-09 (Figure 8.9); role permission matrix	Pending (PA3)	Pending (PA4)
FR-03	UC-01 (logout)	Log out button in the header on signed-in screens	Pending (PA3)	Pending (PA4)
FR-04	UC-05; inputs UC-03/04	S-04, with availability and staffing inputs from S-10 and S-11	Pending (PA3)	Pending (PA4)
FR-05	UC-05; input UC-03	S-04; S-10 (Figure 8.9)	Pending (PA3)	Pending (PA4)
FR-06	UC-05	S-04	Pending (PA3)	Pending (PA4)
FR-07	UC-05	S-04	Pending (PA3)	Pending (PA4)
FR-08	UC-05; UC-06	S-04; S-02; S-05. The interactive demo has no dedicated version history screen.	Pending (PA3)	Pending (PA4)
FR-09	UC-06	S-02; S-05; S-08	Pending (PA3)	Pending (PA4)
FR-10	UC-06; UC-07	S-02; S-03	Pending (PA3)	Pending (PA4)
FR-11	UC-07	S-03	Pending (PA3)	Pending (PA4)
FR-12	UC-06; UC-07	S-05; S-08	Pending (PA3)	Pending (PA4)
FR-13	UC-08	S-06	Pending (PA3)	Pending (PA4)
FR-14	UC-09	S-07	Pending (PA3)	Pending (PA4)

FR-15	UC-09	S-07	Pending (PA3)	Pending (PA4)
FR-16	UC-09	S-07	Pending (PA3)	Pending (PA4)
FR-17	UC-09	S-06	Pending (PA3)	Pending (PA4)
FR-18	UC-09; downstream UC-06/07	S-07; S-06; S-02; S-03	Pending (PA3)	Pending (PA4)
FR-19	UC-02	S-09 (Figure 8.9)	Pending (PA3)	Pending (PA4)
FR-20	UC-03	S-10 (Figure 8.9)	Pending (PA3)	Pending (PA4)
FR-21	UC-04	S-11 (Figure 8.9)	Pending (PA3)	Pending (PA4)
FR-22	UC-03, UC-04, UC-05, UC-09	S-10/S-11 wireframes; S-04/S-06/S-07 updates planned	Pending (PA3)	Pending (PA4)

Traceability status: The matrix maps FR-01–FR-22 to the corresponding use case specifications and currently identified UI screens. Section 5.4 records the FR-to-AC mappings for AC-01–AC-49. Figure 8.9 provides wireframes for S-09–S-11. The deployed prototype includes S-09–S-11. Peer verification of their workflows is pending. Cross-cutting authorization requirements are additionally governed by the role permission rules in Section 5.3. Design elements and test case identifiers remain pending the PA3 and PA4 phases and must not be interpreted as completed deliverables.

10 Requirements Review

Project: ShiftManager

Review Coordinator: Nguyễn Huy Hoàng

Requirements Review Status — Approved

Formal Review Date: 30 October 2026

Meeting Channel: Microsoft Teams

Present: Nguyễn Huy Hoàng; Võ Quang Minh

Absent with notice: Lê Hoàng Phúc.

Absent: Nguyễn Cao Trí.

Review Objective:

The requirements review evaluates the consistency, completeness, feasibility, and traceability of the ShiftManager specification across Functional Requirements, Non-Functional Requirements, Business Rules, Acceptance Criteria, Use Cases, and Prototype screens. The review records accepted corrections and identifies implementation and evidence follow-up for later phases.

#	Issue found	Requirement(s)	Resolution
1	Initial UC-05 described selection among multiple production teams, beyond PA1 single-team scope.	UC-05; F02	Draft revised to operate on the Manager's assigned team.
2	UC-05 originally omitted the case where the Manager refuses shortage-warning confirmation.	UC-05; FR-07	Draft now keeps the current official version unchanged if confirmation is declined;
3	Planned future schedule revisions and operational mid-shift	UC-05; UC-09	Draft distinguishes UC-05 (planned future revisions) from UC-09

	replacement have different timing rules.		(absence-related ongoing-shift coverage).
4	UC-09 must clarify how a Manager records a no-show independently of Employee absence reporting.	UC-08; UC-09; FR-14	UC-09 defines the Manager no-show flow. UC-08 and UC-09 reuse the same unresolved coverage request when one exists.
5	The MVP rule allowing only one active Pending or Accepted replacement offer per uncovered position requires approval.	UC-09; FR-16	FR-16, BR-06, UC-09, AC-21, and AC-35 apply the same one-active-offer rule.
6	Mid-shift replacement effective time and handling of historical assignments must be validated.	UC-09; FR-18	FR-18, UC-09, and AC-41 specify prospective coverage for the remaining valid interval and preserve earlier history.
7	FR-05 defines fairness over "5 preceding weeks" while UC-05 and agreed scope use the current week plus four preceding weeks.	FR-05; UC-05	FR-05, BR-04, and UC-05 use the target week and up to four prior official weeks.
8	UC-02 account/role management, UC-03 availability submission and UC-04 staffing-template configuration lack clearly corresponding dedicated FR statements.	Section 5.1; UC-02/03/04	FR-19, FR-20, and FR-21 map to UC-02, UC-03, and UC-04 in Sections 6.3 and 9. Figure 8.9 shows the wireframes. The deployed prototype includes all three screens. Peer verification is pending.
9	UC-08 specification fields are shifted: use case name, description, primary actor and priority do not occupy the correct template rows.	Section 7.8; UC-08	Corrected the UC-08 name, description, actors, priority, and requirement references. Aligned reporting for future and ongoing shifts with FR-13 and BR-08.
10	UC-03 rejects marking all shifts Unavailable by invoking an unspecified minimum-availability policy.	UC-03; F02	Aligned UC-03 with BR-02 and AC-28. An all-Unavailable declaration triggers a warning and can be saved after confirmation. Final decision. During the availability submission window, employees may mark all shifts in a week as Unavailable. The system warns of potential

			understaffing and saves the declaration only after the employee confirms. If the employee goes back, the previous declaration stays in effect. Employees use Report Absence (UC-08) if they cannot work an already published assignment.
11	Section 8 lists only S-01–S-08 and has no dedicated mockup mapped to UC-02, UC-03 or UC-04.	UC-02/03/04; Section 8	Linked the deployed prototype for S-01–S-11 and added wireframes for S-09–S-11 in Figure 8.9. The document also specifies overtime updates to S-04, S-06 and S-07. Verify those workflows against the deployed prototype.
12	The UC-07 subsection heading is duplicated as "Use Case UC-06" even though its table is UC-07.	Section 7.7	Corrected the Section 7.7 heading to Use Case UC-07.
13	NFR and Business Rule tables have been populated, but some measurement conditions, policy sources, and cross-section references remain inconsistent.	Sections 5.2–5.3	Sections 5.2 and 5.3 document the measurement methods and rule sources. Validate measurements and cross-section alignment during PA3 and PA4.
14	UC-02 duplicates the staffing configuration workflow instead of specifying Employee Account Management.	UC-02; UC-04; FR-19	UC-02 specifies team-scoped Employee Account Management under FR-19. UC-04 covers staffing configuration.
15	Working-time and overtime checks require explicit compliance rules and sufficient supporting records.	BR-03; BR-07; UC-05; UC-09	FR-22 and BR-07 specify posting, voluntary sign-up, approval, required records and validation. Validate the applicable limits and demo input records before testing overtime scenarios.
16	FR, AC, and Screen identifiers require synchronization following the retention of UC-02 and the proposed additional requirements.	FR-19–FR-22, AC-24–AC-49, S-04/S-06/S-09–S-11	Sections 5.4 and 9 record the requirement mappings. Verify the deployed workflows for S-09–S-11 and the overtime updates against the relevant acceptance criteria.

The 30 October review approved the PA2 requirements review. The 3 November integration check covered FR-01–FR-22, UC-01–UC-09 and AC-01–AC-49. FR-22 and AC-44–AC-49 were integrated after the meeting to document voluntary overtime and overtime coverage.

Review Meeting Record

Date: 30 October 2026

Meeting Channel: Microsoft Teams

Participants: Nguyễn Huy Hoàng and Võ Quang Minh

Absent with notice: Lê Hoàng Phúc.

Absent: Nguyễn Cao Trí.

Confirmed Review Direction:

1. Retain UC-02 as Manage User Accounts and Roles and correct its specification.
2. Resolve inconsistencies in the scheduling, staffing, absence, and replacement workflows.
3. Align Functional Requirements, Business Rules, Acceptance Criteria, Use Cases, and Prototype references.
4. Record the agreed corrections and track later changes for PA3.

Implementation and evidence follow-up

1. Record the 3 November integration changes in the PA3 requirements change log.
2. Confirmation of any newly proposed numerical or operational policies.
3. Test the deployed workflows for S-09–S-11 and verify the overtime and mid-shift prototype updates.
4. Team verification of the FR–UC–AC–Screen mappings, including S-09–S-11 in the deployed prototype.
5. Section 1 links Issues #6 and #7 for the SRS contributions and PR #4 for the interactive prototype.

Prototype verification and demo input records remain follow-up for later phases.

11 AI Usage Log

#	Member	Tool	Task / purpose	How the output was verified and changed
1	Nguyễn Huy Hoàng	ChatGPT	Discussed and refined the business requirements of ShiftManager, including automatic scheduling, employee availability, staffing requirements, fairness, and shortage handling.	Reviewed the proposed workflows through interactive discussions. Clarified hard and soft constraints, selected a five-week fairness window, and decided that partially covered schedules require explicit Manager confirmation before publication.
			Assisted in drafting UC-05 (Generate and Publish Weekly Schedule) and UC-09 (Arrange Replacement Coverage), including main, alternative, and exception flows.	Evaluated alternative workflows and clarified the boundaries between planned schedule revisions and absence-related replacements. Selected manual draft adjustments with hard-constraint validation and version-based updates for published schedules. The use case drafts were reviewed against the final requirements in Section 10.
			Assisted in preparing preliminary findings for Section 10 – Requirements Review and checking the consistency of UC-05 and UC-09.	Challenged findings that depended on unfinished use cases, particularly UC-06. The preliminary issue list was reduced and revised to distinguish observable specification gaps from assumptions requiring team confirmation. This entry describes the preliminary review. See Section 10 for the subsequent meeting record and approval status.
			Developed Given/When/Then acceptance criteria for F01–F04 (AC-01–AC-23).	Mapped every draft acceptance scenario to the FR-01–FR-18 statements present in the team file and to UC-01–UC-09. These are proposed test cases for PA4; no runtime test has been claimed.

			Constructed Section 9 requirements traceability for existing FR-01–FR-18.	Matched FR IDs to the actual use case list and screen IDs S-01–S-08. Marked missing screens and PA3/PA4 columns as pending rather than inventing designs or test-case IDs.
			Revised draft review findings and documented the team-reported redistribution of PA2 responsibilities.	Checked against the current working SRS, separated open issues from previously revised text, recorded the group decision of 34%/33%/33%/0%, and left GitHub evidence and formal team-review outcomes unverified.
			Assisted in reviewing the PA2 Software Requirements Specification and preparing a structured correction guide for FR/BR consistency, UC-05 scheduling constraints, UC-09 replacement lifecycle, acceptance criteria, requirements traceability, and the 30 October 2026 requirements review record.	Compared the proposed corrections against the current PA2 document, retained UC-02 as an account-management use case, distinguished confirmed decisions from proposed requirements, and incorporated the actual meeting attendance reported by the coordinator. Additional FRs, Business Rules, acceptance scenarios, and interface references remain subject to team review and integration verification. No implementation tests or final requirements baseline verification were performed as part of preparing the correction guide.
2	Võ Quang Minh	Claude	Drafted Section 4 (Stakeholders) from the PA1 Project Proposal: stakeholder list, roles, and main interests.	Checked the stakeholders and roles against the PA1 Vision section. Shortened the descriptions, kept the Boss as a read-only role, and added the Development team and Lecturer/TA as non-user stakeholders. The interests are marked as assumptions to be validated with real users.
			Drafted the use case specifications for UC-01	Compared the format with UC-05 and UC-09 written by Hoàng.

			(Log In), UC-06 (View Published Schedule and Changes), and UC-07 (Acknowledge Schedule Update), including main, alternative, and exception flows.	Checked that the roles, version rules, and acknowledgement rules are consistent with PA1 features F01 and F03. Renamed the duplicated UC-06 heading in the template to UC-07. Section 7 records the final names of UC-06 and UC-07.
			Created the 8 low-fidelity wireframes (S-01 to S-08), the screen list, the screen flow table, and the callout descriptions for Section 8.	Checked that each screen maps to a use case step. Reviewed the layout and the text on each wireframe, and corrected overflowing buttons and table alignment. All data on the screens is fictional.
			Built a clickable HTML prototype of the 8 screens and ran an accessibility check (colour contrast, labels, keyboard use, 375 px layout) with automated scripts.	Opened the prototype and tried the main flows for each role. Compared the automated results (lowest contrast ratio 5.1:1, no unlabeled controls, keyboard-only completion of the main flows) with the table in Section 8.4, and fixed two issues found (missing page language and lost focus after an action).
			Drafted Section 5.2 (Non-Functional Requirements) with measurable requirements and verification methods.	Reduced the list from 12 to 5 items and replaced the verification methods with simple checks the team can do by hand or with Chrome tools. Removed the session timeout and lockout rule. The values (3 s, 30 s, 30 employees) are assumptions to be confirmed by the team.

			Drafted an early staffing configuration use case, then labeled UC-02 with main, alternative, and exception flows, business rules, and non-functional constraints.	The final SRS assigns staffing configuration to UC-04 and account management to UC-02. Rewrote the non-functional constraints to match the final NFR-01, NFR-03, NFR-04, and NFR-05.
3	Lê Hoàng Phúc	Gemini	Brainstormed and drafted Section 2 (Introduction: Purpose, Product Scope, Glossary) and Section 3 (Problem Statement, Operating Environment, Design & Implementation Constraints).	Evaluated the AI-generated draft against the revised factory production team scope from the 19 October TA consultation. Manually refined the MVP boundaries for features F01–F04, explicitly excluded out-of-scope enterprise functions (payroll, time-tracking, multi-department management), created a bilingual (English/Vietnamese) glossary, and adjusted operating constraints to strictly match React, Node.js/Express, and Supabase architecture.
			Formulated and structured Section 5.1 Functional Requirements (FR-01 to FR-18) mapped to core features F01–F04, and Section 6 Use Case Model (Actors, Diagram, and List).	Verified that all 18 functional requirements directly map to the 4 Must-have core features without scope creep. Rewrote ambiguous AI outputs into precise, testable *"The system shall..."* statements. Built the Mermaid syntax for the Use Case diagram, checked actor boundaries (Employee, Manager, Boss, Supabase Auth), and assigned explicit priority levels aligned with team planning.

			Developed Use Case specifications for UC-03 (*Submit Availability and Preferences*) and UC-08 (*Report Absence*), including main, alternative, and exception flows.	Analyzed availability submission cutoff logic and policy exception handling for UC-03. Fixed template formatting errors in UC-08 (corrected shifted table fields), ensured that submitting an absence report does not automatically revoke shift assignment until replacement confirmation in UC-09, and added mandatory server-side JWT claim verification steps.
			Formulated Section 5.3 Domain Requirements & Business Rules (BR-01 to BR-06) and UC-04 (*Configure Staffing Requirements*) taken over following team task reallocation.	Strictly refined BR-01 through BR-06 by mapping every business rule to updated external legal references (Vietnam Labor Code No. 45/2019/QH14 Articles 105, 107 and 109) and international standards